



GE Medical Systems

gemedical.com

Technical Publication

Direction 5117715-100

Revision 4

GE Medical Systems

LightSpeed 5.X Plus Pre-Installation

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- Failure to heed this Warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

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- Não tente reparar o equipamento sem ter consultado e compreendido este Manual de Assistência Técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a' choques elétricos, mecânicos ou outros.

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- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione esterno alla GEMS richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Non tenere conto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

このサービスマニュアルには英語版しかありません。
GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
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注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

Lithium Battery Cautionary Statements



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
1	08/18/04	Initial release document.
2	12/10/04	Updated System Option Catalog Numbers.
3	07/22/05	PSR13033111: Updated Fig. 4-10 and Fig. 4-11, using international unit.
4	10/08/05	PSR13050339: added Note 4.

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Preface

Publication Conventions

Section 1.0 Safety & Hazard Information

1.1 Text and Character Representation

This publication uses various paragraph and character styles to indicate potential hazards. Key words "danger," "warning," "caution" and "notice" are used to identify important safety information. Text formatting (e.g., bold, all-caps, etc.) is applied to the paragraph contents to further highlight safety statements. Safety information normally identified in these hazard statements includes:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100).



DANGER



USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**



WARNING



USED WHEN A HAZARD EXISTS THAT COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **POTENTIAL FOR UN-COMMANDED MOTION**



CAUTION



Used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Sharp objects



NOTICE Equipment Damage Possible

Used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It is important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information is always preceded by an exclamation point contained within a triangle, . Additionally, several different icons (symbols) may be used to point out specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Conventions Used For General Information

2.1 Document Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

Examples of prefixes used for general information:

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

Note: Conveys information that should be considered important to the reader.

2.2 Page Layout

Headers and footers in this publication are designed to allow you to quickly identify your location. The document’s part number and revision number appear in every header on every page. Odd numbered page footers indicate the current chapter and page number. Even page footers show the current section and page number.

Chapter 1

Introduction

This direction contains physical and electrical data necessary for planning and preparing a site. Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Medical Systems does not provide or install wires, conduits, junction boxes, and ducts as illustrated in this publication.

All CT site plans, preliminary concepts and final working drawings must be reviewed by General Electric Headquarters Architectural Planning prior to construction or approval.

Contact your local General Electric sales representative for complete information regarding your site-specific room layout.

Section 1.0

Site Readiness

The following **MUST** be completed before installation work can begin for a GE CT scanner:

- Finished wall, ceiling, floors & millwork
- Active phone line and network connection
- Power available to A1, w/provision for LOTO



NOTICE An improperly prepared site (i.e., one that is in a state of construction) can result in *increased installation time*.

A CT scanner installed in a dirty environment is more prone to contamination, which can result in *decreased reliability* and *increased scanner downtime*.

Section 2.0

Responsibility of Purchaser

“Pre-installation” is work necessary to plan and prepare a site for installation of equipment. Avoid delay, confusion, and waste of manpower. Complete the following pre-installation work:

- Procure required materials.
- Install required material before delivery of system components.
- Complete all alterations and modifications if not specifically included in the sales contract.
- Complete the Pre-installation Check List in [Chapter 2](#).

Section 3.0

Site Preparation Prior to Equipment Delivery

- Determine room dimensions and verify entry doors are large enough to bring scanner system through.
- Install table/gantry floor duct.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- Install conduit, duct, and raceway.
- Install power supply of correct voltage output and adequate KVA rating. Install local disconnects, including proper over-current protection.
- Install “Unistrut” or other suitable support work for mounting equipment on walls or from ceiling.
- Camera should be on-site at the time of installation.

Chapter 2

Pre-Installation Overview

Before a LightSpeed 5.X system can be installed, all pre-installation requirements must be complete.

Section 1.0

Site Ready for Installation

Site-specific items must be verified before the installation can begin.

1.1 Dust/Dirt Contamination

The LightSpeed 5.X systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site.



NOTICE Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.

1.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed Series components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

1.3 Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin.
LightSpeed 5.X scanners can only be installed on a 4" concrete floor.

1.4 Broad-band

For information on Broad-band requirements, refer to Chapter 4, [Section 4.0, on page 41](#).

1.5 Phone Line (for optional modem)

Two phone lines must be installed at or near the console and be operational prior to installation.

- 1-Analog line (for modem use)
- 1-Voice line

1.6 Review

The LightSpeed Series systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads. The table has five (5) pads used for leveling it.

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the template (P/N 2363851) shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the 2 floor templates.
- 2.) Start with the Gantry template—align per the GE print.
- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Check the level of the floor (See [Figure 2-1](#)) across the templates.

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

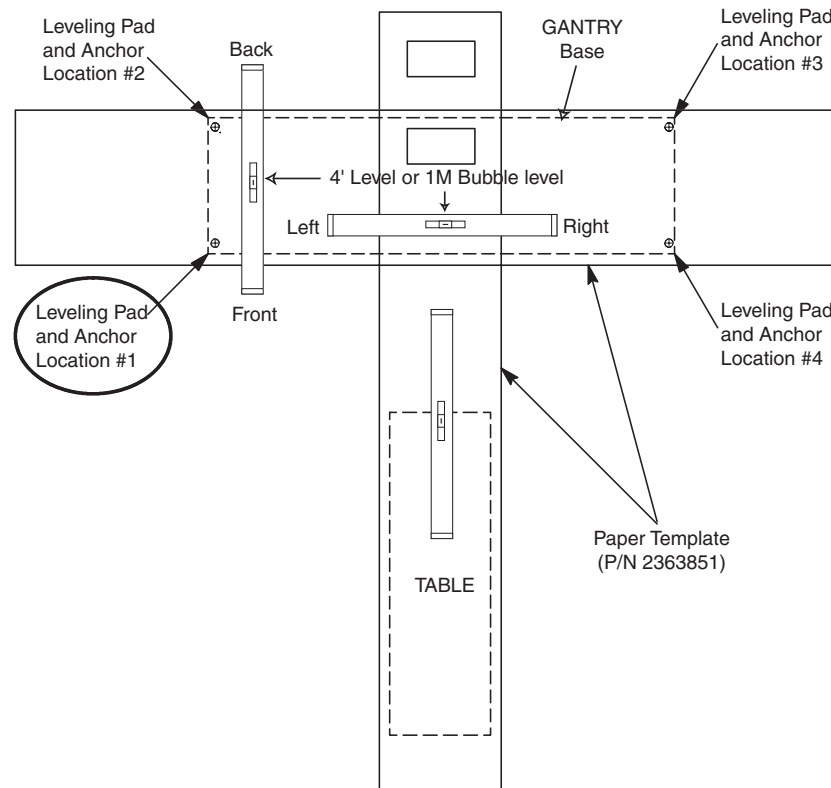


Figure 2-1 Hole Locations

Section 2.0

Pre-Installation Checklist

GE	CUST	DATES
Y N	Y N	
	☐ ☐	Project schedule verified with contractor, facilities department, and GE?
	☐ ☐	Can you meet the committed site ready date?
☐ ☐	☐ ☐	Construction completion date matches delivery date?
☐ ☐	☐ ☐	Power & Ground survey complete, date _____ Hospital contact _____
	☐ ☐	Delivery date is scheduled for: _____
	☐ ☐	First Use date is scheduled for: _____
☐ ☐	☐ ☐	Does the delivery and / or installation date need to be adjusted?
☐ ☐	☐ ☐	Applications dates: On Site scheduled for: _____, Education Center Training: _____
	☐ ☐	Person assigned to review and verify that all installation requirements are met?

GE	CUST	GENERAL / SITE REQUIREMENTS
Y N	Y N	
		<i>Must be completed 5 week prior to the delivery</i>
☐ ☐	☐ ☐	Final drawings distributed to the contractors and reviewed with GE?
	☐ ☐	Have the Radiologist Health Physician review the room layout.
	☐ ☐	Final print(s) "signed off" that approve equipment layout / orientation?
	☐ ☐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 2-1, below)
	☐ ☐	Has the responsibility of cabling, installing, interfacing accessories not on the order been discussed? (Refer to service price pages for GE's support of 3rd party accessories)
	☐ ☐	All 3 rd party vendors identified, notified and scheduled? (i.e., Netcom, 3M, Kodak, Medrad, etc.)
	☐ ☐	Will existing network, modem, and camera cable drops reach new locations / requirements/ function with LightSpeed 5.0?
	☐ ☐	Have the following requirements in Figure 2-1 reviewed by your contractor

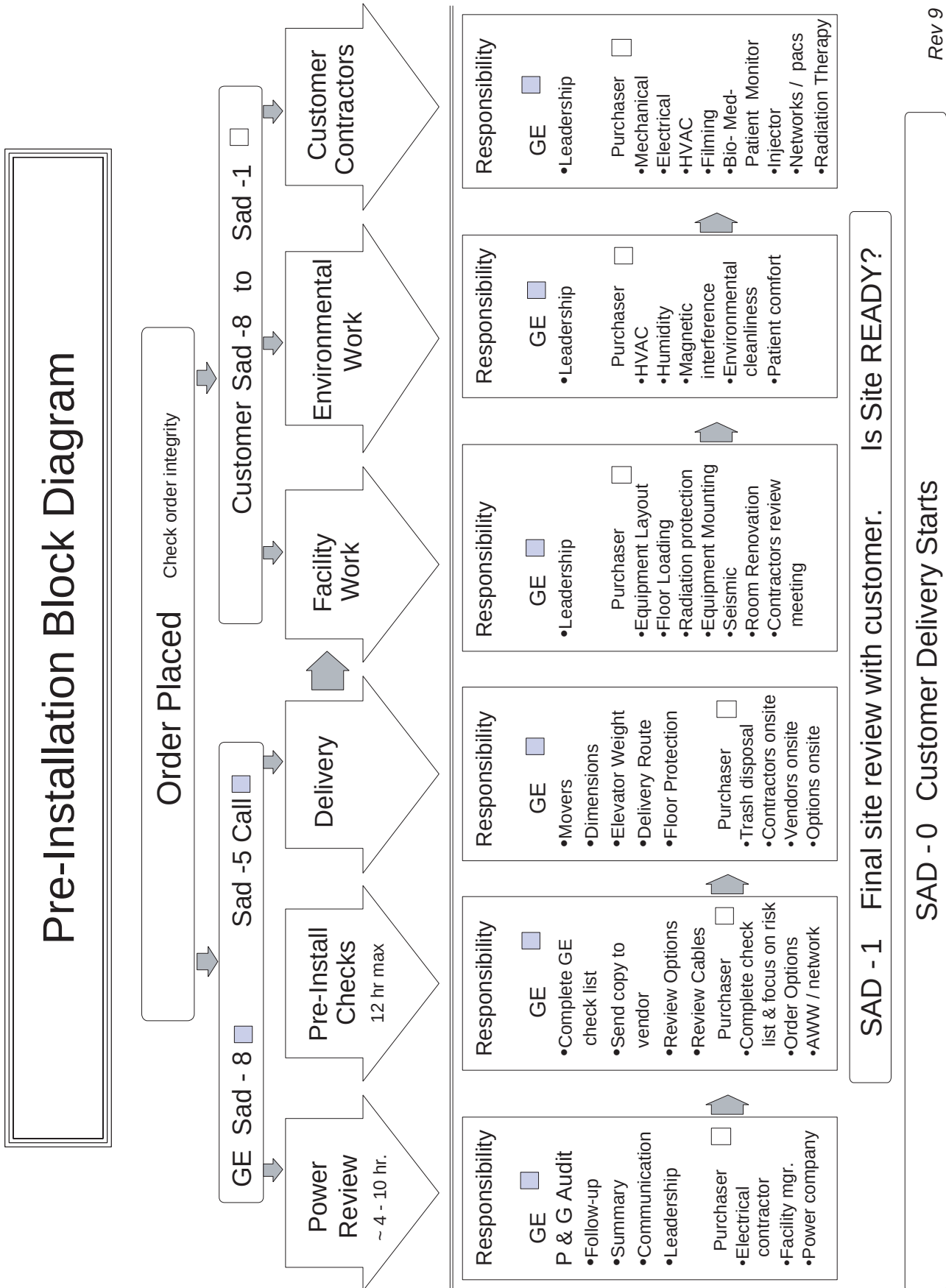
Requirements?	
• Air Conditioning	• Structural
• Electrical	• Radiation Protection
All work by contractors must be completed prior to GE delivery	

Table 2-1 Specific Site Requirements

GE	CUST	EQUIPMENT
Y N	Y N	
		<i>Must be completed 5 weeks prior to the delivery</i>
☐ ☐	☐ ☐	Order reviewed for completeness and compatibility with existing equipment.
	☐ ☐	Accessories on the order verified for compatibility (i.e. DASMs, cable lengths, laser cameras, AWW work station, injector and other NL items, etc.) Verify these will work with your CT System
	☐ ☐	Interfaces to existing and/or new accessories ordered and planned for accordingly.
	☐ ☐	Has the location of peripherals been determined and planned accordingly with GE representatives? EKG monitor___ Injector control___ Laser camera ___ UPS ___ 2 nd Monitor___
☐ ☐	☐ ☐	Have any additional services to be provided by GE been discussed and agreed upon?
	☐ ☐	Remote monitors ___ AWW relocation ___ Wall mounted accessories ___

GE	CUST	NETWORKING
Y N	Y N	
		<i>Must be completed 5 weeks prior to the delivery</i>
☐ ☐	☐ ☐	Have IP addresses and Host Names been obtained?___ Will a network camera be used?___
☐ ☐	☐ ☐	Mandatory Item: Broad-band installed? ___
☐ ☐	☐ ☐	Optional Item: CT service telephone line identified and installed for the InSite (<i>Electrical, mechanical, etc.</i>)
☐ ☐	☐ ☐	Mandatory Item: Network installed? ___ Network jacks installed and tested?
☐ ☐	☐ ☐	Mandatory Item: Network options ordered ___ HIS RIS option ___ DICOM print ___ AWW ___
	☐ ☐	Test network connections.

GE	CUST	OTHER
Y N	Y N	
		<i>Must be completed prior to the delivery</i>
☐ ☐	☐ ☐	Review: Arrangements made in the schedule to allow for remodeling, if required. (i.e., wall, floor, or ceiling repair work, painting, other cosmetic finishes)
☐ ☐	☐ ☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to reinstall?
☐ ☐	☐ ☐	Is de-installation required? No___ Yes___ CT removal date _____
☐ ☐	☐ ☐	Delivery route identified, and verified with the proper hospital personnel? _____
☐ ☐	☐ ☐	Will acceptance testing or Bio-Medical testing be required?
☐ ☐	☐ ☐	Trash bins available for the removal of papers/ boxes/ etc. during the installation.



Chapter 3

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring site planning work for the LightSpeed 5.0 system. Contact your local GE Medical Systems Sales representative for a complete list of all system options or visit us at <http://www.gemedical.com>. Refer to the instruction manuals supplied with specific options for respective details.

CATALOG NUMBER	OPTION DESCRIPTION
Monitor B71292CA B71712CA	LCD Monitor CRT Monitor
International Dolly Set B7850LD	For International customers, if dollies are required.
SmartScore Pro Acquisition SW (H2H3H16) B7850PL	(Software)
Connect Pro SW Only B7500LN	ConnectPro Option provides a direct interface to HIS/RIS
Bar Code Reader B7540RB	LightSpeed 5.0 Bar Code Reader
Long Cable Set B7816PH	
Modem B71272CA	Global Modem Kit

Table 3-1 System Options Catalog (Part) Numbers

Section 2.0

Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator's console (not shown).



Figure 3-1 Base Scanner System

Chapter 4

Room Planning

Section 1.0 Recommended Layouts

Consult your local GE Sales and Service Representative about your specific needs.

Figure 4-1, Figure 4-2 and Figure 4-3 show three of many possible room layouts. For exact equipment sizes, refer to Section 2.0. Remember, sufficient space must be maintained around equipment for full operation, service and safety.

Note:
Other room
arrangements
are possible.

Cable length is an important consideration in room layout. LightSpeed 5.X system is shipped with standard length cables, with a set of longer cables (B7816PH) available as an option. See the electrical page of the GE print for your specific requirements. (The layout shown in Figure 4-2 is for standard length cables.) Note, also, that where possible, the cables should enter the gantry from below.

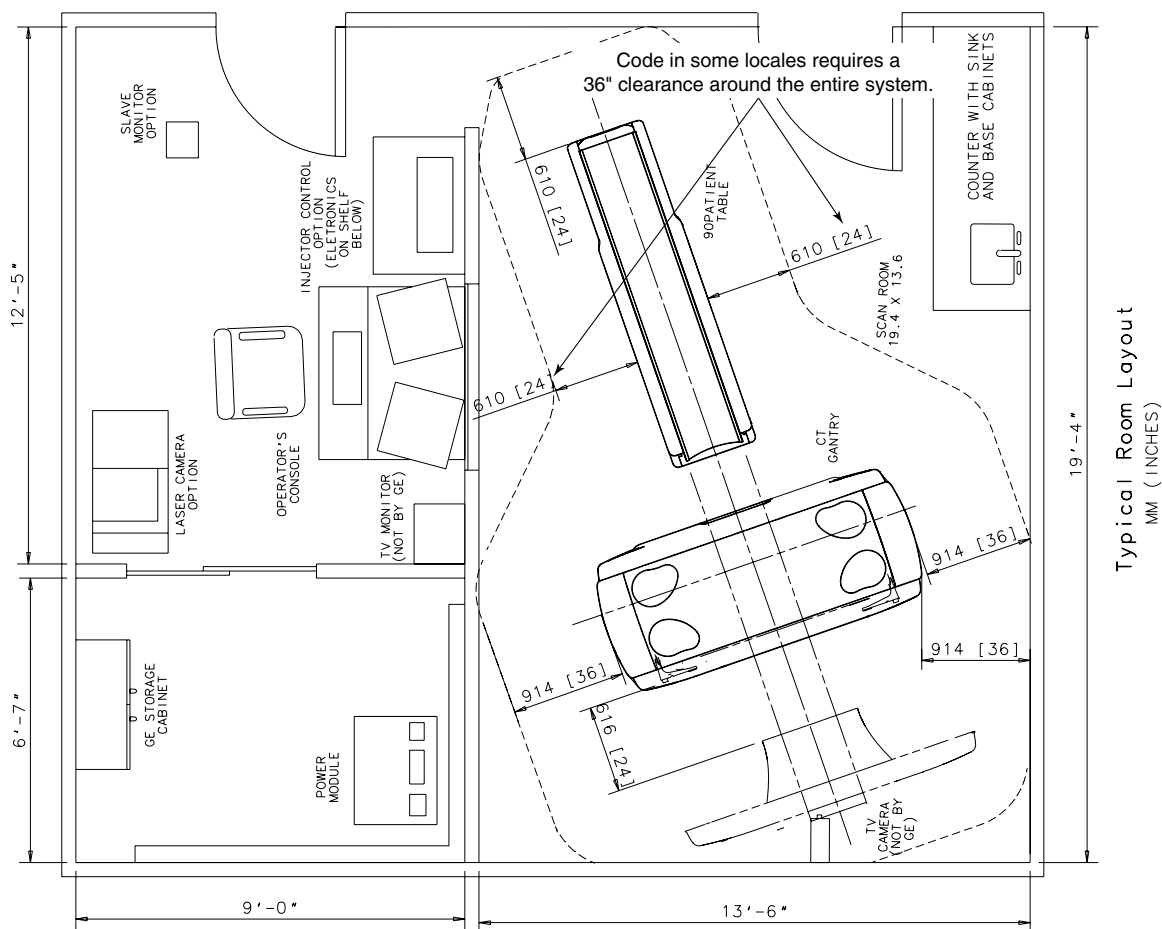


Figure 4-1 Typical Room Layout (with minimum table & gantry service clearances)

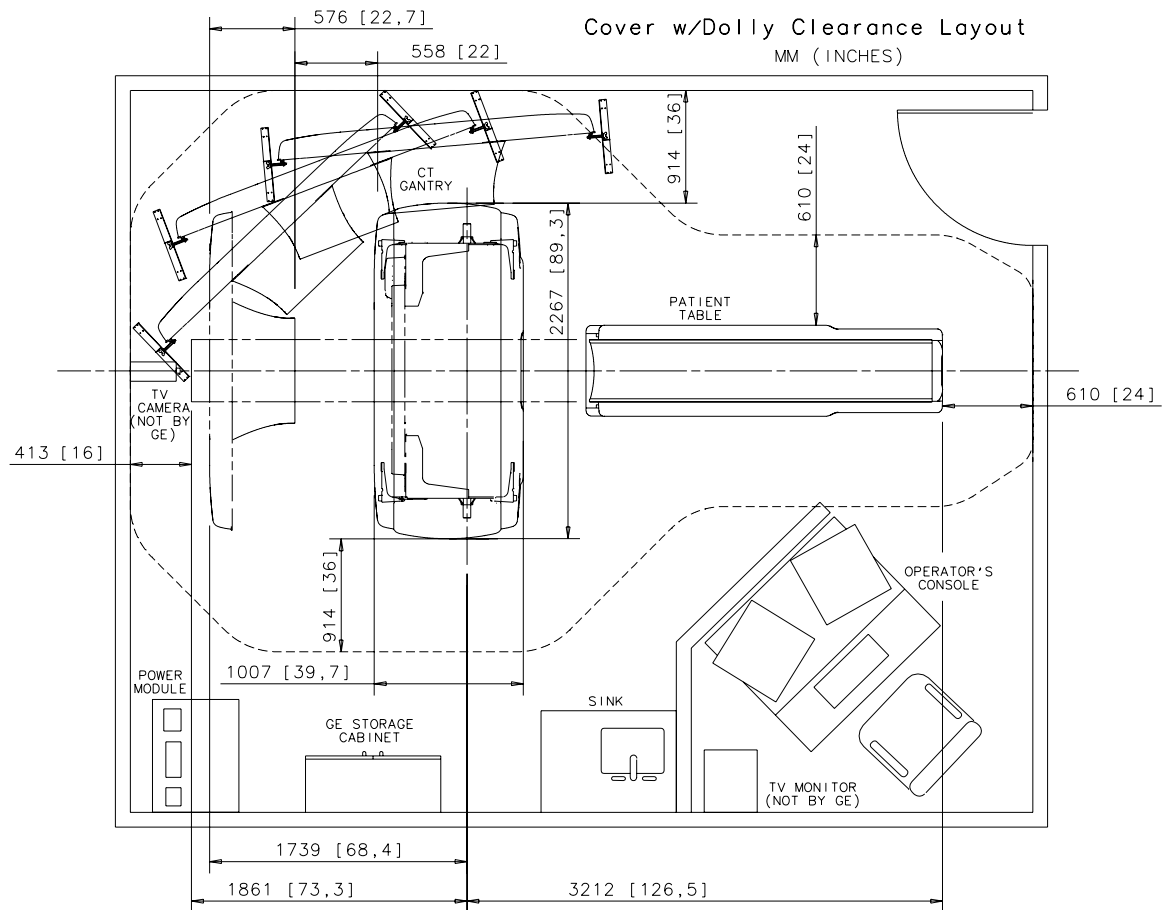


Figure 4-2 Sample Room Layout, Cover Removal Clearances

COMMON DIMENSIONS AND CLEARANCES

System Service

- Back cover removal (rear clearance) 36.0" (913.5mm)
- Front/Back cover removal (side clearance) 36.0" (913.5mm)
- Gantry service side to obstruction 36.0" (913.5mm)
- Area around the PDU 42.0" (1067mm)
- Area around the A1 breaker box 42.0" (1067mm)
- Area between the table down foot end to obstruction 36.0" (913.5mm)

Options

Ceiling Pedestal mount lowest point to floor Injector or Monitor 96.0" (2438.5mm)

System Operation

- Finished ceiling to floor 108.0" (2743mm)
- Table max extension head end with extender from Center Line 80.0" (2030.0mm)
- Table extension head end with extender to obstruction 6.0" (152.0mm)
- Table in lowest position w/cradle at home position to Center Line 126.5" (3209.0mm)
- Back of Console to wall 6.0" (152.0mm)
- Back of PDU to wall 6.0" (152.0mm)

Typical Room Dimensions

- Scan Room 13.6' X 22.0' (4.1m X 6.60m)
- Control Room 9.0' X 22.0' (2.74m X 6.60m)

Additional dimensions are available in [Figure 4-4](#) through [Figure 4-8](#) of this document. Consult your local General Electric Installation Specialist for your appropriate room specifications.

1.1 Injector Control

A suitable work area, which is within reach of the operator's console, should be provided for placement of the injector control. Refer to [Figure 4-3](#).

Wall mounted, ceiling mounted and pedestal units need cables to be routed from the gantry area to the console area. The supplied cable is 50 feet in length. Injectors require AC power. Mounts are available in different configurations and lengths. Refer to Injector documentation for detailed installation instructions. Refer to [Figure 4-3](#) for injector placement information.

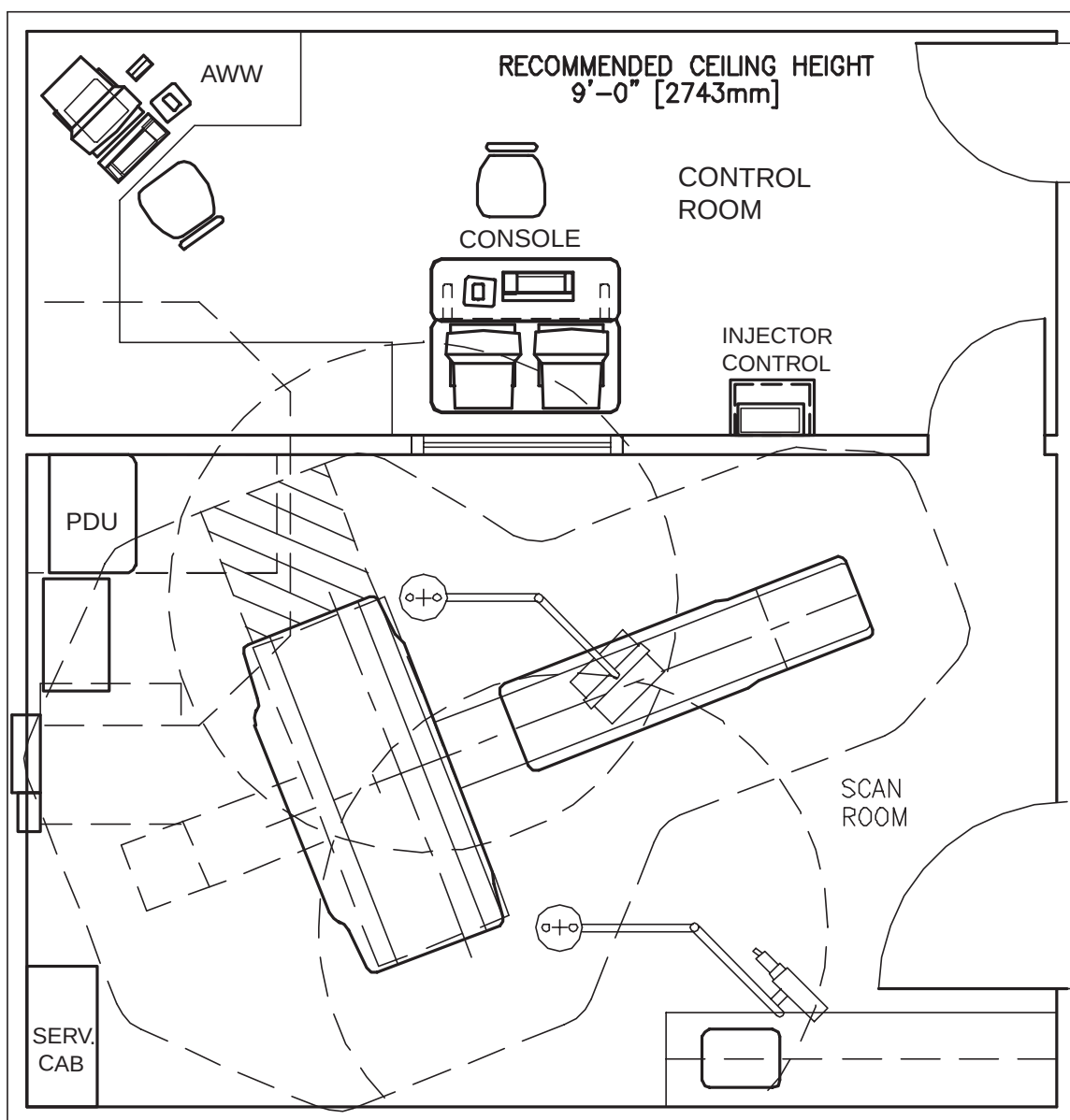


Figure 4-3 Typical Room Layout, with Injector Option

1.2 Storage Cabinet

A storage cabinet is provided by GEMS to store all supplied service equipment (see [Table 4-1](#) for equipment list). This storage cabinet (24" D x 26" W x 42" H) should be located in the scan room suite area, for easy service access. Refer to [Figure 4-2](#) or [Figure 4-3](#).

ITEM	SIZE	WEIGHT (TOTAL)
QA Phantom (water filled)	20 x 15 cm	12 lb
35CM Phantom	35 x 7 cm	18 lb
48CM Phantom	48 x 7 cm	25 lb
Phantom Holder	25 x 25 cm	8 lb
FE Box (Purple)	30 x 38 x 30 cm	15 lb
Rear Cover Dollies	158 x 82 cm	25 lb
Front Cover Dollies	85 x 20 cm and 85 x 15 cm	35 lb
Install Support Kit (box)	30 x 30 x 38 cm	20 lb
Tube Hoist Kit	77 x 8 cm and 38 x 15 cm	20 lb

Table 4-1 Equipment to be stored in storage cabinet

1.3 Advantage Windows Workstation (AWW)

Refer to Pre-Installation Manual 2111833 and Installation/Service Manual 2111831. Refer to [Figure 4-3](#) for sample layout of AWW option.

Section 2.0 Component Dimensions

DESCRIPTION	WIDTH		DEPTH		HEIGHT	
	INCH	MM	INCH	MM	INCH	MM
System						
Gantry	86.6	2200	39.4	1000	74.6	1895
Table	24.3	617	94	2387		
Power Distribution Unit (NGPDU)	28	711	22	559	42	1067
Operator's console/computer	48	1219	39	991	33.5	851
Remote Color Monitor						
Color printer	23	584	18	457	7	178
Advantage Windows						
Workstation	16.5	419	16.1	409	3.1	79
Monitor	19.5	495	19	483	18.6	471

Table 4-2 Dimensions of Components

2.1 Table and Gantry

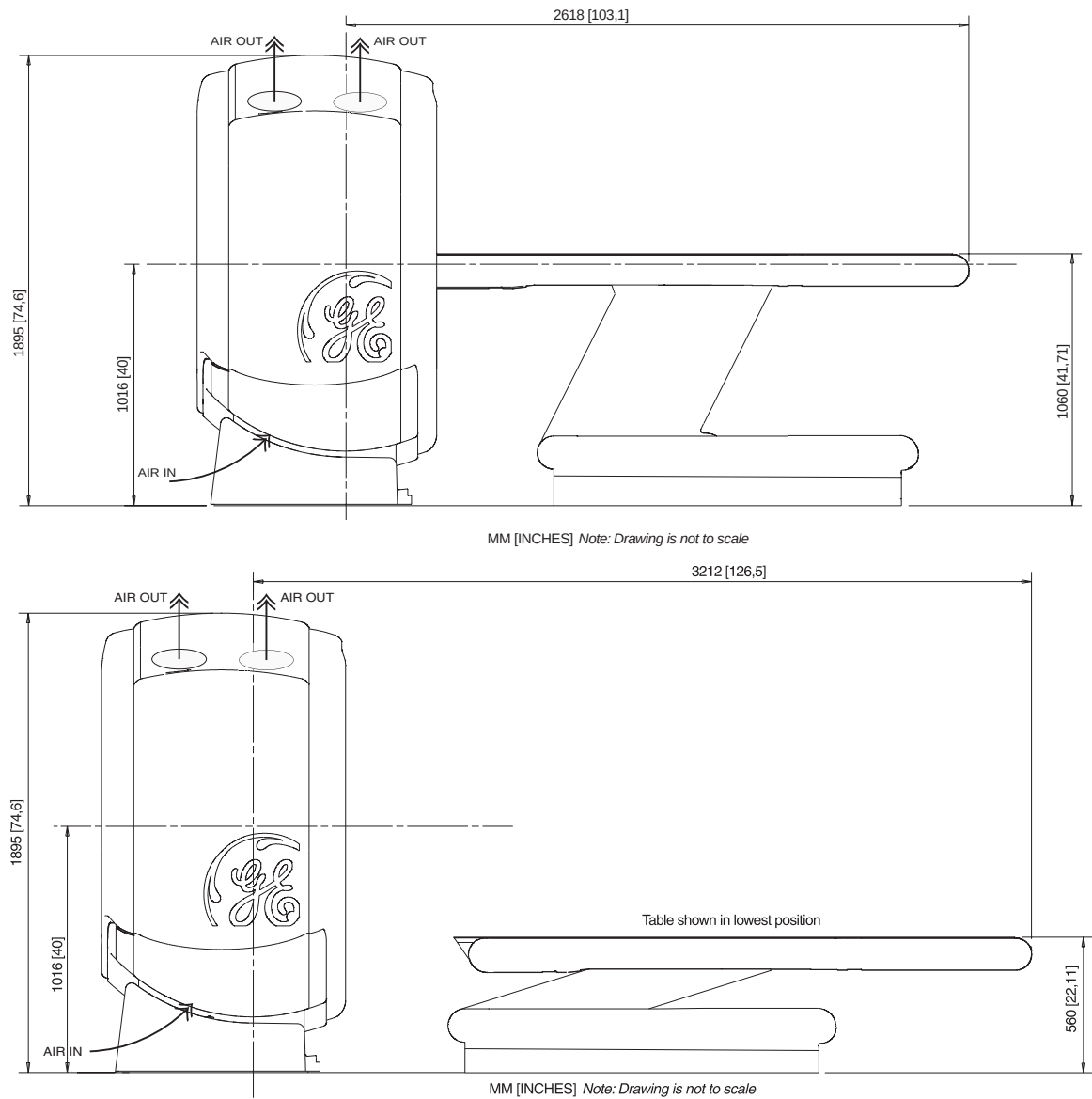


Figure 4-4 Table and Gantry (Side View)

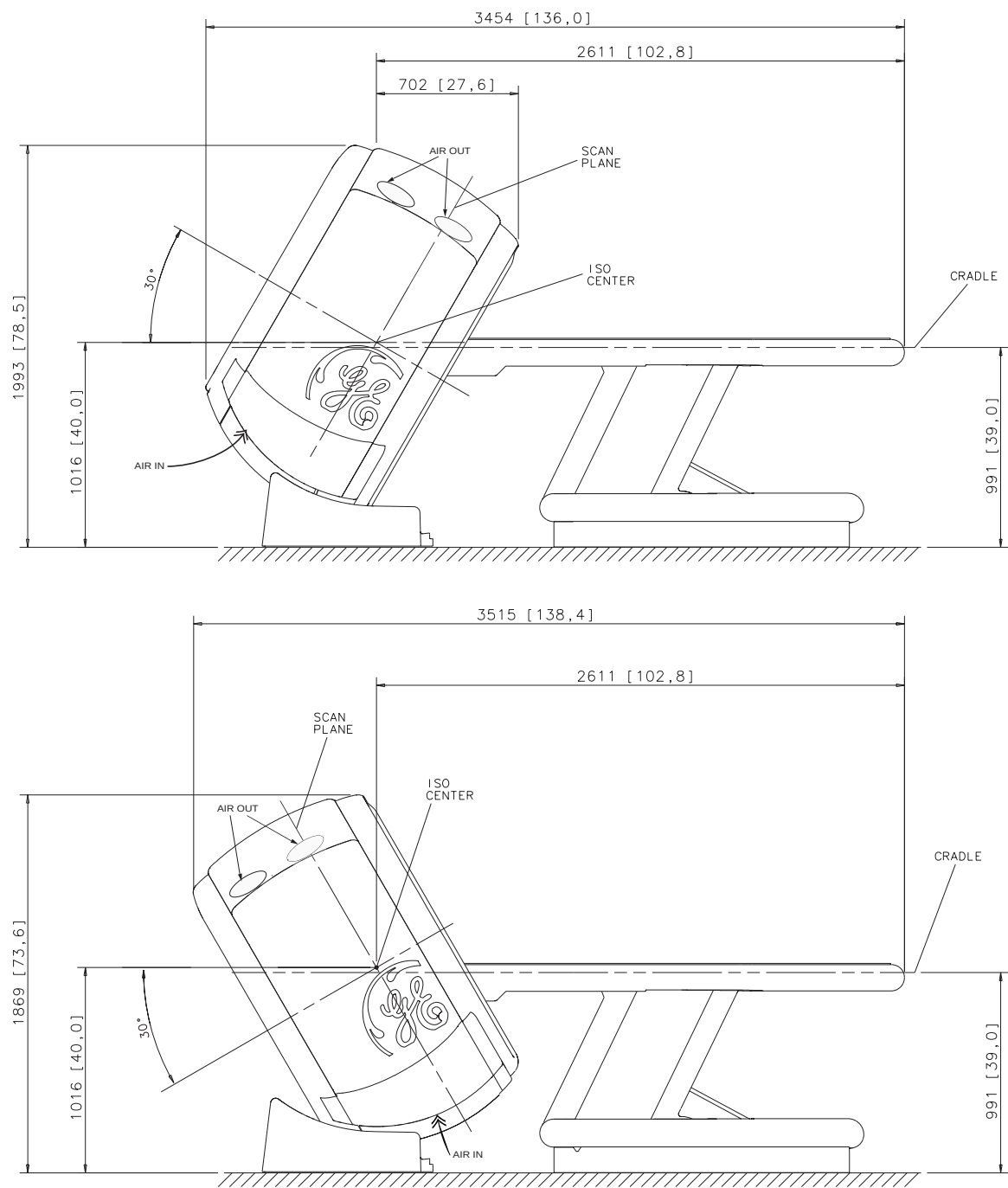


Figure 4-5 Gantry shown tilted +30° (top) and -30° (bottom)

2.2 Power Distribution Unit

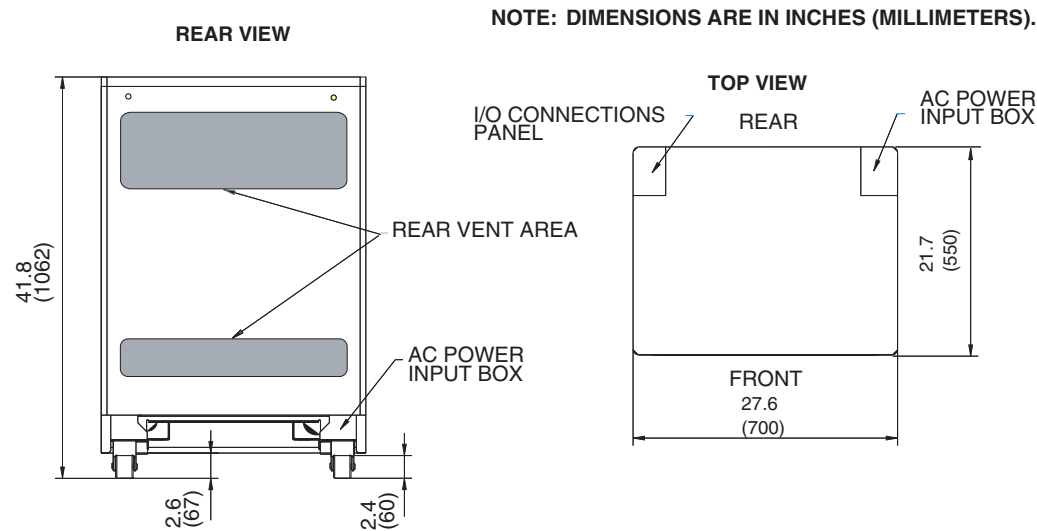


Figure 4-6 Power Distribution Unit (NGPDU)

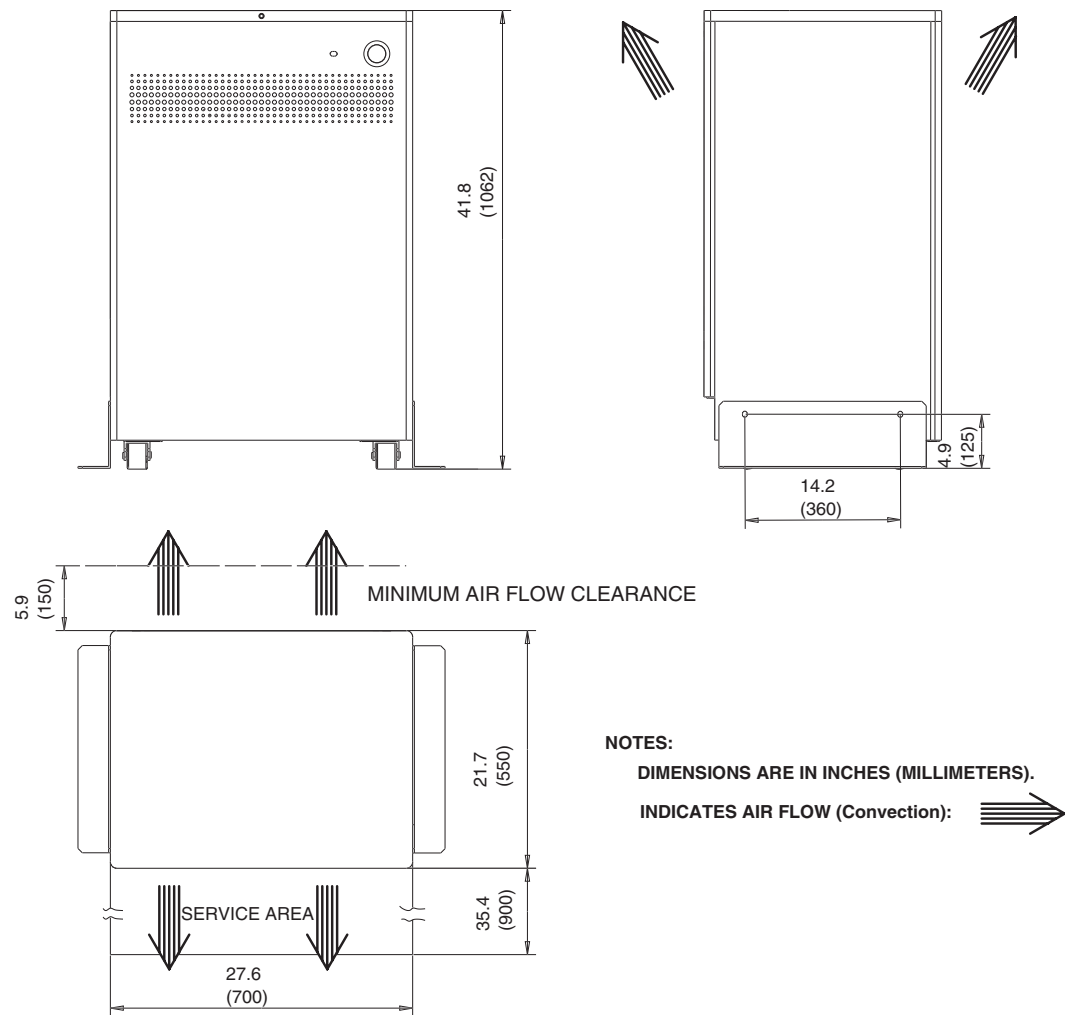


Figure 4-7 Power Distribution Unit (NGPDU)

2.3 Operator's Console

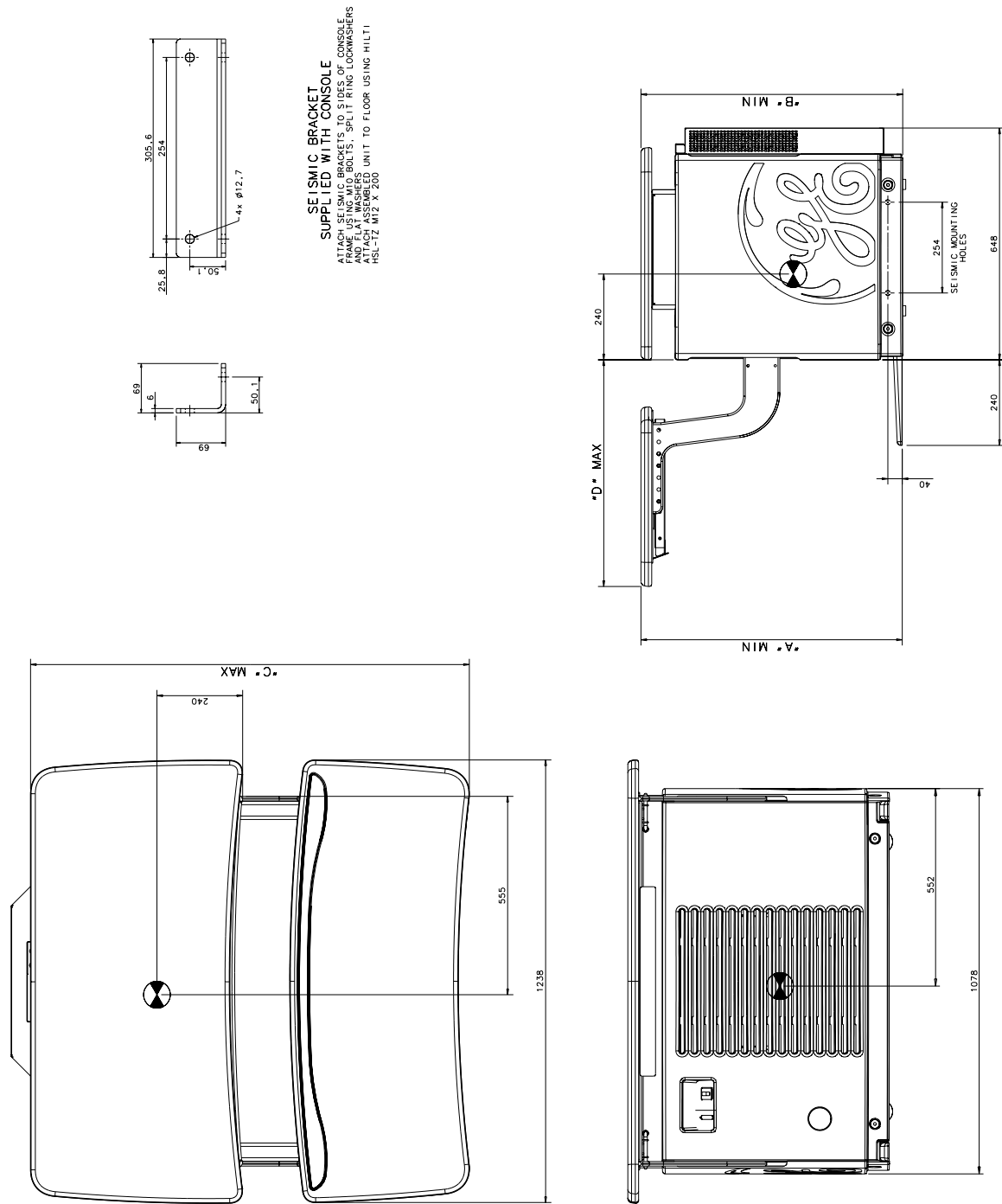


Figure 4-8 Operator's Console

Section 3.0 Structural Requirements

3.1 Suggested Ceiling Heights

A ceiling height of 108" (2743mm) is recommended. A ceiling height of 90" (2286mm) is the absolute minimum acceptable height for gantry installation.

3.2 Table and Gantry Mounting Requirements

Note: **It is the purchaser's responsibility to provide an approved support structure and mounting method. General Electric is not responsible for any failure of the support structure or method of anchoring.**



NOTICE **LightSpeed 5.X scanners should be installed on 4" minimum thickness dense concrete floors. All other floor types must be approved by GEMS Engineering.**

Table and gantry mounting dimensions are shown in [Figure 6-2](#), [Figure 6-3](#), [Figure 6-4](#). Refer to Chapter 6 for additional details of floor loadings, component weights, and Gantry and Table installation and anchoring.

Anchor gantry and table to floor by a means that will maintain their relative alignment and meet applicable building and other local codes, including seismic structural mounting requirements.

Floor structure must be capable of withstanding the occupied weight of table and gantry, and the individual contact area loading of these components. **The table-gantry requires a floor support in the 100 lbs/sq ft. (490 Kg/sq m) class or above. Any installation on a floor with a rating less than 100 lbs./sq. ft. (490 Kg/m²) should be braced to bring it up to this requirement.** Localized bracing to support the concentrated loads at the floor contact sections should also be provided.

Support areas of the patient table and gantry must rest on solid concrete, not resilient tile or carpeting which will slowly yield over a period of time and disturb alignment of table to gantry.

Factors that could cause misalignment between gantry and table due to floor sag should be considered. The cradle can potentially carry a 450 lb (205 kilogram) patient. Center of gravity changes as cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to Chapter 6 for gantry and table mounting details.

No part of floor surface within table and gantry, nor the two interface areas between table and gantry, should be higher than the support area for table and gantry.

A qualified person must verify that the site and method of anchoring are adequate to support loads and maintain table-to-gantry alignment. Location of supporting beams and columns may dictate position of table-to-gantry assembly. Use of flush floor duct or conduit in the floor may significantly affect floor strength.

The method and placement of anchoring through bolts must not reduce structural strength of floor.

3.3 Floor Strength

Concrete floors must have a minimum strength of $f'_c = 2000$ psi (1.4×10^7 Pa) at 28 days for mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength. Your GE Service representative can assist you.

Consult GE Installation Support Services for further details.

3.4 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 0.3125 (8mm) between depression and high spots over any 120 in. (3048 mm) distance within the area of the gantry and the area around the table (see the envelope shown in [Figure 4-2](#) or [Figure 4-3](#)).

3.5 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customers responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits, However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

3.5.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

3.5.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak.

3.5.3 Equipment Location

To minimize the interference, the CT equipment should be placed on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Roadways
- Subways
- Trains
- Hallways
- Elevators
- Heliports
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units

3.6 Walls: Scan Window

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position (see [Figure 4-2](#) and [Figure 4-3](#)).

Note: The operator at the Operator Workspace must be able to view the patient during a scan.

3.7 Mobile Units

It is ultimately the responsibility of the customer/architect/engineer to design the site solution (i.e., levelness, vibration, equipment location and proximity effects of surrounding equipment, such as other mobile, transportable and relocatable units that could contain CT, MR, or other medical devices).

Section 4.0

Network Connections

Broad-band is considered the standard network connection for LightSpeed 5.X. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the Console.

- A patch cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 9-3](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.

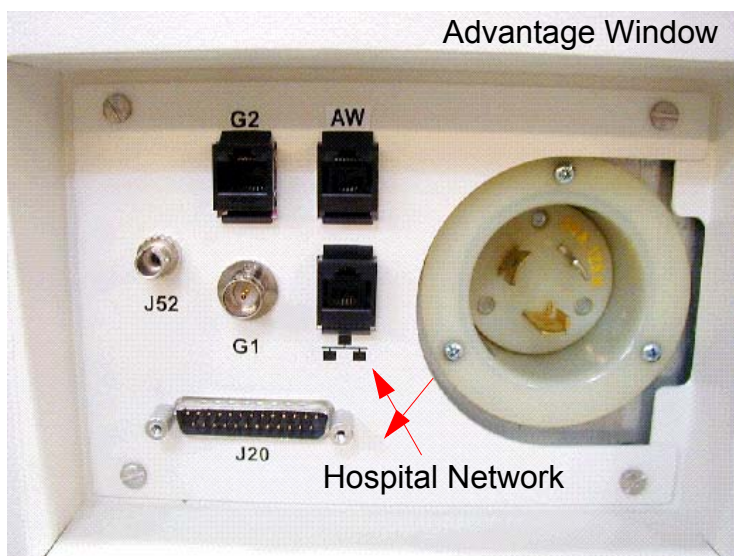


Figure 4-9 Console Rear Bulkhead

4.1 US Process Overview

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

4.1.1 Customer Broad-Band Responsibilities

Provide GEMS Installation Specialist with an accurate site address, telephone number, contact name, and email address for the:

- Customer Champion
 - Co-ordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broad-band infrastructure meets GEMS requirements for connection as determined by a mutual assessment with the GEMS Connectivity team.
- IT Contact
 - Complete an equipment assessment with GEMS Connectivity team to determine site readiness for broad-band
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
City, State: _____ FE Phone: _____
Date: _____ FE Email: _____

- | | Yes | No |
|---|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls
☐ b) Cisco Routers
☐ c) Cisco 3000 Series (Altiga acquisition)
☐ d) Checkpoint Firewalls Software Version 4.1 and higher
☐ e) Nortel Contivity Software Version 3.2 or higher
☐ f) Redcreek | ☐ g) Symantec (Raptor) firewalls
☐ h) Firebox
☐ i) Linux S/WAN
☐ j) Sidewinder
☐ k) Netscreen
☐ l) None
☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

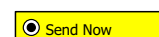
Once you have completed both pages of this form, please send it to:

a) Judy Heyer judy.hey@med.ge.com
b) Judy Heyer Fax# 414-918-4707

[Use the send button on page 2](#)



System ID List & Compatibility Matrix



Section 5.0

Radiation Protection

Figure 4-10 depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and a 20 cm water phantom (head) with the technique shown.

Note:
Actual
measurements
can vary.

All measurements have an accuracy of $\pm 20\%$ because of measurement equipment, technique, and system-to-system variation.
Use the correction factors shown in Figure 4-3 to adjust exposure levels to the usual scan technique at your site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 3.75mm images	0.82
4 X 2.5 LD Fluro 5mm	0.59
4 x 1.25 LD 5mm (1i) Fluro 2.5 mm	0.40
1 x 1.25mm images	0.20
1 x 1.25	0.10

Table 4-3 Shielding Requirements Scaling

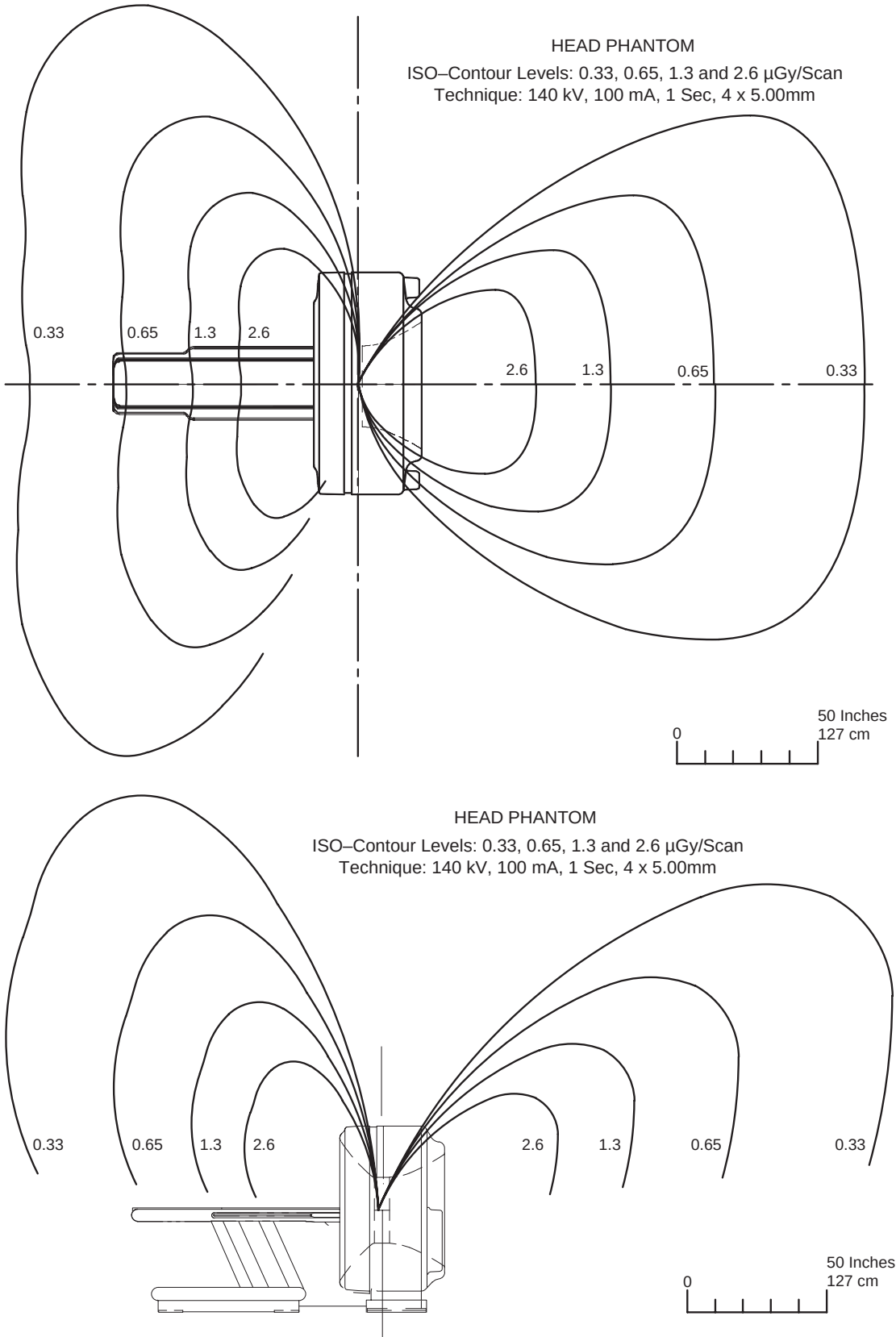


Figure 4-10 Typical Scatter Survey (Head Filter)

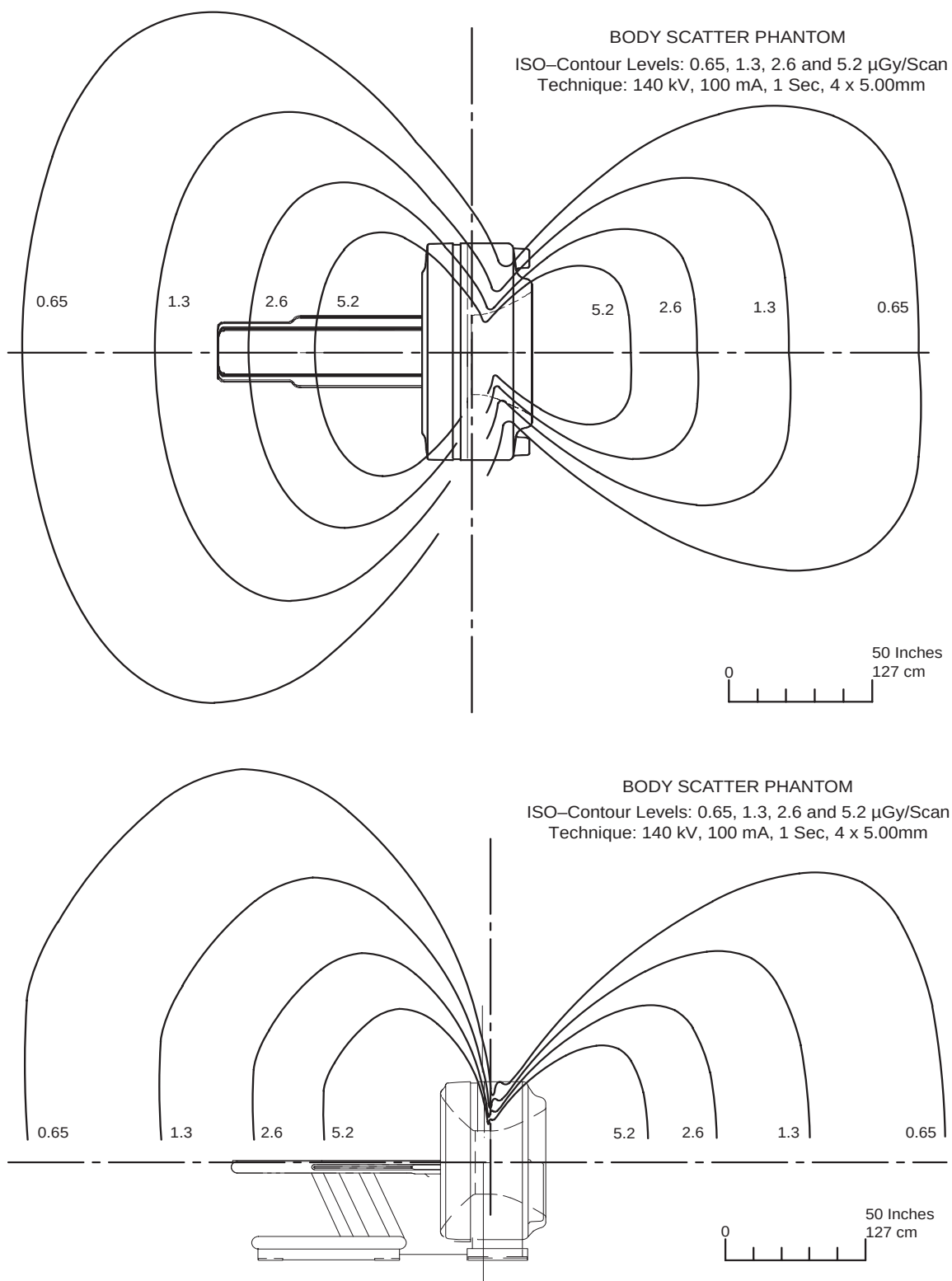


Figure 4-11 Typical Scatter Survey (Body Filter)

Chapter 5

Environmental Conditions

Ratings and duty cycles of CT subsystems apply if site environment meets the standards of this section. Maintain environmental conditions listed below at all times – including, for example, overnight, weekends and holidays. Shut down the CT system if air conditioning is not working. When system is shut down for major repair, air conditioning may be shut down also.

Section 1.0

Temperature and Humidity Specifications



NOTICE
Potential
Equipment
Failure

Do not operate (i.e., “Power ON”) Gantry or Console subsystems with ambient room temperatures exceeding 75° F. Scan room or control room temperatures in excess of 75° F can result in the failure of gantry or console components.

- Ambient Temperature: (Fahrenheit and Celsius)
 - Scan Room: Maintain a temperature of 72° F $\pm$ 3° F (22° C), for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 70° - 75° F (21° - 24° C).
 - Control Room (including Console/Computer): Maintain 72° F (22° C).
 - Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 60° - 75° F (15° - 24° C).
- Store media (cartridges) in long-term storage in same temperature range as host computer.
- Store media in the host computer environment for one-half hour before use.
- Maintain relative humidity of 30%-60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.).
- The maximum relative humidity rate of change is 5% RH/hr.

Section 2.0

Temperature and Humidity Monitoring

Locate computing subsystems in an area that has the environment specified in [Section 1.0](#), above. First, assess your environment's heat and humidity. If necessary, temporarily install a temperature and humidity recorder close to where the gantry will be installed. Note readings before installation and again after installation to verify true temperature and humidity conditions for your environment. Consider your HVAC needs and redundancy. You may wish to consider an air conditioner with two compressor units rather than one. A backup (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 3.0

Cooling Requirement

Use [Table 5-1](#) to assist in cooling requirements planning. Over half the cooling used by your scanner is required for gantry operation. Locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort. Do not locate any cooling vents directly above the gantry. Air returns above the gantry are recommended.

SYSTEM COMPONENT	BTU/HR	WATT
1. Gantry recommended (See NOTE 1)	25,100	7,350
2. Table	700	200
3. Power distribution unit	3400	1000
Recommended Scan Room Subtotal (see notes):	29,200	8,550
4. Operator's console/computer with one IG	7361	2165
Additional IG, each	1360	400
Monitor, each (CRT)	2346	690
SCSI Tower	425	125
Injector	425	125
DASM	425	125
Recommended Control Room Subtotal (includes OC w/1 IG, 2 monitors & SCSI Tower):	12,342	3630
System Total (Recommended) (See NOTE 1)	41,542	12,180
Option: Remote Color Monitor	1178	345
Option: Advantage Windows	256	75
ROOM TOTAL (SEE NOTE 2)		
NOTE 1: With 75 scan rotations per patient: Recommended BTU/hr. provides for up to six patients per hour. It is also needed during calibration of the system.		
NOTE 2: Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.		

Table 5-1 Cooling Requirements (Worksheet)

Please refer to [Figure 4-4](#), [Figure 4-7](#) and [Figure 4-8](#) in [Chapter 4, Section 2.0](#), for component air flow requirements.

Section 4.0

Altitude

System operating altitude is from mean sea level to 10,000 ft. (3050 meters).

Section 5.0

Electro-Magnetic Interference (EMI)

5.1 Gantry and Table

Locate gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.



WARNING REQUISITES FOR ROOM

GANTRY & TABLE ARE ALLOWED TO BE INSTALLED ONLY IN X-RAY PROTECTED ROOMS, WHICH PROVIDE AN ATTENUATION OF AT LEAST 12dB FOR RADIO DISTURBANCES FROM 30 MHz TO 1 GHz.

(The Warning statement above is specified by IEC 60601-1-2, First Edition, 1993)

5.2 Color Monitor

Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla (100 milligauss) to guarantee color purity and display geometry. See [Figure 5-1](#).

5.3 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See [Figure 5-1](#).

5.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

5.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., console or computer equipment - the UPS is not classified as sensitive electronics) within one-and-a-half meters (1.5m) of the Power Distribution Unit., in any direction (including above or below). See [Figure 5-1](#).

5.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Medical Systems Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

5.9 EMC Edition 2 Compliance

5.9.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 EMC standard for medical electrical equipment.

The LightSpeed Plus System is suitable to be used in the electromagnetic environment, as per the limits & recommendations described in your technical reference manual.

Adhering to the distance separation recommended in tables found in the technical reference manual, between 150KHz & 2.5GHz, will reduce disturbances recorded at the image level but may not eliminate all disturbances. However, when installed and operated as specified herein, the system will maintain its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

(*)For example, a 1W mobile phone (800MHz to 2.5GHz carrier frequency) shall be put 23 meters apart from the LightSpeed Plus (in order to avoid image interference risks).

5.9.2 USE LIMITATION: External Components

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMAGNETIC COMPATIBILITY of the EQUIPMENT and/or SYSTEM

5.9.3 INSTALLATION REQUIREMENTS & ENVIRONMENT CONTROL

In order to minimize interference risks, the following requirements shall apply.

CABLE SHIELDING & GROUNDING

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

THIS PRODUCT COMPLIES THE RADIATED EMISSION AS PER CISPR11 GROUP1 CLASS A STANDARD LIMITS.

The LightSpeed Plus is predominantly intended for use, in non-domestic environments, and not directly connected to the Public Mains Network.

The LightSpeed Plus is predominantly case of using in a domestic environment (e.g. doctors' offices),

in order to avoid interferences, it is recommended to use a separated AC power distribution panel & line, with a X-ray shielded room.

SUBSYSTEM & ACCESSORIES POWER SUPPLY DISTRIBUTION

All components, accessories subsystems, systems which are electrically connected to the LightSpeed Plus, must have all AC power supplied by the same power distribution panel & line.

STACKED COMPONENTS & EQUIPMENT

The LightSpeed Plus should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is necessary, the LightSpeed Plus should be observed in order to verify normal operation

in the configuration in which it will be used.

LOW FREQUENCY MAGNETIC FIELD

In case of a digital LightSpeed Plus, the Gantry (digital detector) shall be apart 1meter from the generator cabinet, and 1meter apart from the analog (CRT) monitors. These distance specifications will minimize the low frequency magnetic field interference risk.

STATIC MAGNETIC FIELD LIMITS

In order to avoid interference on the LightSpeed Plus system, static field limits from the surrounding environment are specified.

Static field is specified less than <1 Gauss in Examination room, and in the Control Area.

Static field is specified less than <3 Gauss in the Technical Room.

ELECTROSTATIC DISCHARGE ENVIRONMENT & RECOMMENDATIONS

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup. The relative humidity shall be at least 30 percent.

The dissipative material shall be connected to the system ground reference, if applicable.

Chapter 6

Floor Loading and Weights

Section 1.0 Floor Loading

The LightSpeed 5.X system has a total floor load of approximately 6500 lbs (2950 kg). About 175 lbs (2350 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly.

Table 6-1 lists CT components with weight, size, floor loading and normal mounting requirements.

ITEM	NET WEIGHT LB(KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry (no covers)	~1700	86.6 X 39.4 (2200 X 1000)	140 (682)	Rectangular base plate 24 X 81 (610 X 2057) with four round pads, each 2.5 (63.5) in contact with floor. Individual pad loadings are: 910 lb, 960 lb, 1040 lb, and 1090 lb (see Figure 6-3).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Dollies (each)	250 (114)				
Top Cover (each)	24.6 (11.2)				
Side Cover (each)	25 (11.3)				
Front Cover	95 (43)				
Rear Cover	100 (45)				
Patient Table	1185 (538) Includes 450 (204) Patient	24.3 X 94 (617 X 2387)	175 (852)	Rectangular base 17 X 57.5 (432 X 1460) with five round pads, each 2.5 (63.5) in contact with floor. Worst-case cantilever pull on any bolt is 1235 lb (561 kg).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at three leveling pads into concrete floor.
Dolly	215 (98)				
Power Distribution Unit	~770 (~350)	28 X 22 (711 X 559)	185 (909)	Four Casters support area of 28 X 22 (711 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Console w/HP & w/o monitors	450 (204)	48 X 39 (1219x991)	140 (681)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Console may be anchored to floor using angle brackets ²
Monitor - CRT (ea)	80 (36)				
Monitor - LCD (ea)	22 (10)				

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in 3.2 - Table and Gantry Mounting Requirements, on page 39.]
- 2.) Angle brackets are included on the shipping skid, and are also available in the OPTIONAL Seismic Kit (R4390JC).

Table 6-1 LightSpeed 5.0 System Floor Loading

Section 2.0

Mounting Data, Including Seismic

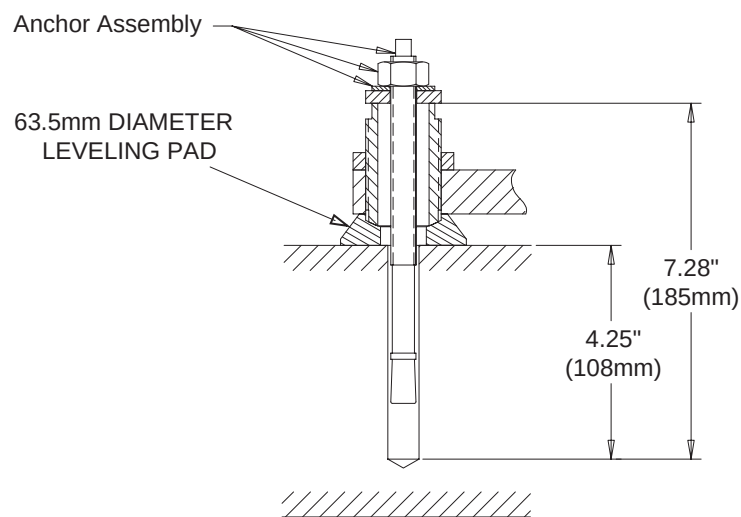
The following pages show center-of-gravity information for system components:

- Gantry: [Figure 6-2](#)
- Table: [Figure 6-4](#)
- Power Distribution Unit: [Figure 6-5](#)
- Operator's Console/Computer: [Figure 6-6](#)

Floor mounting hole locations for components that don't have templates are also in this section.

An optional Seismic Kit (R4390JC) is available, for facilities in areas that require one.

Customer is responsible for seismic mounting. Refer to all applicable codes in your area.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-1 Typical Floor Anchor, Gantry and Table

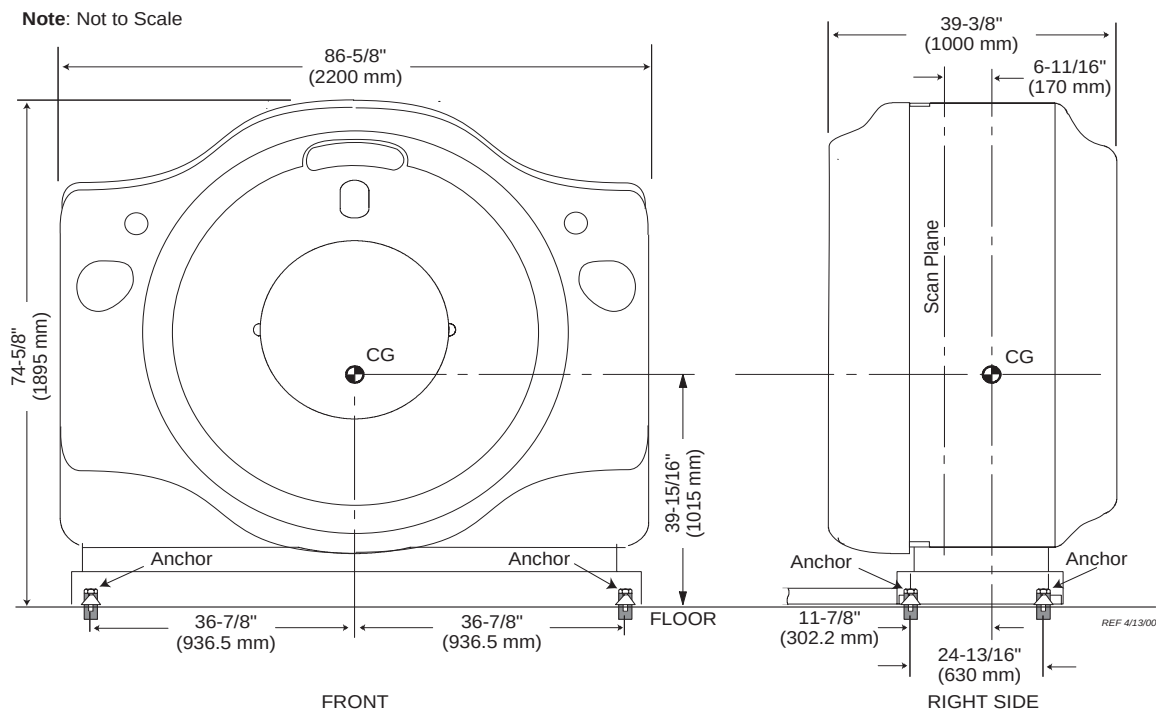
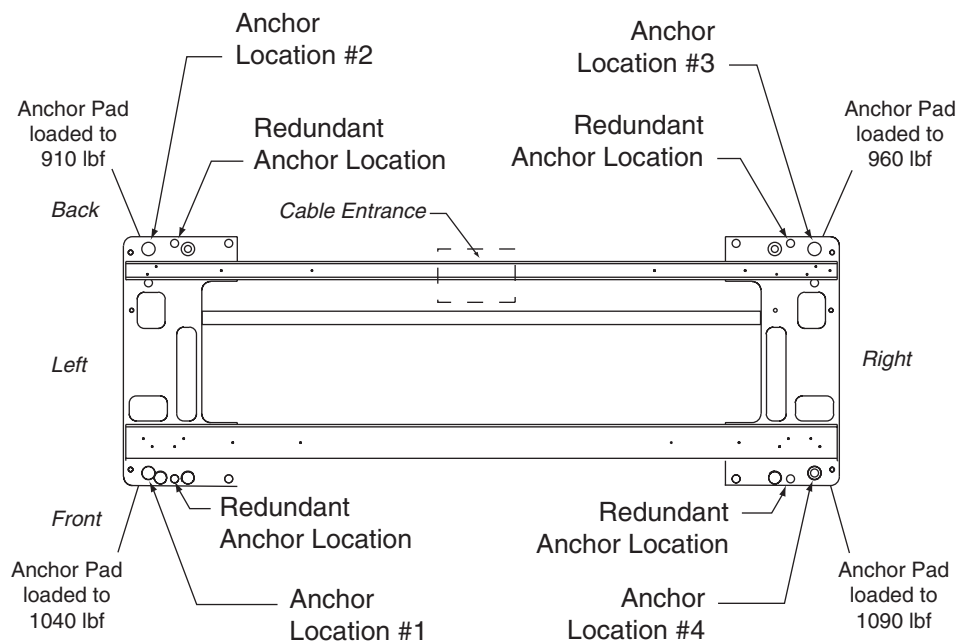


Figure 6-2 Gantry (CT2)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-3 Gantry Anchor Locations

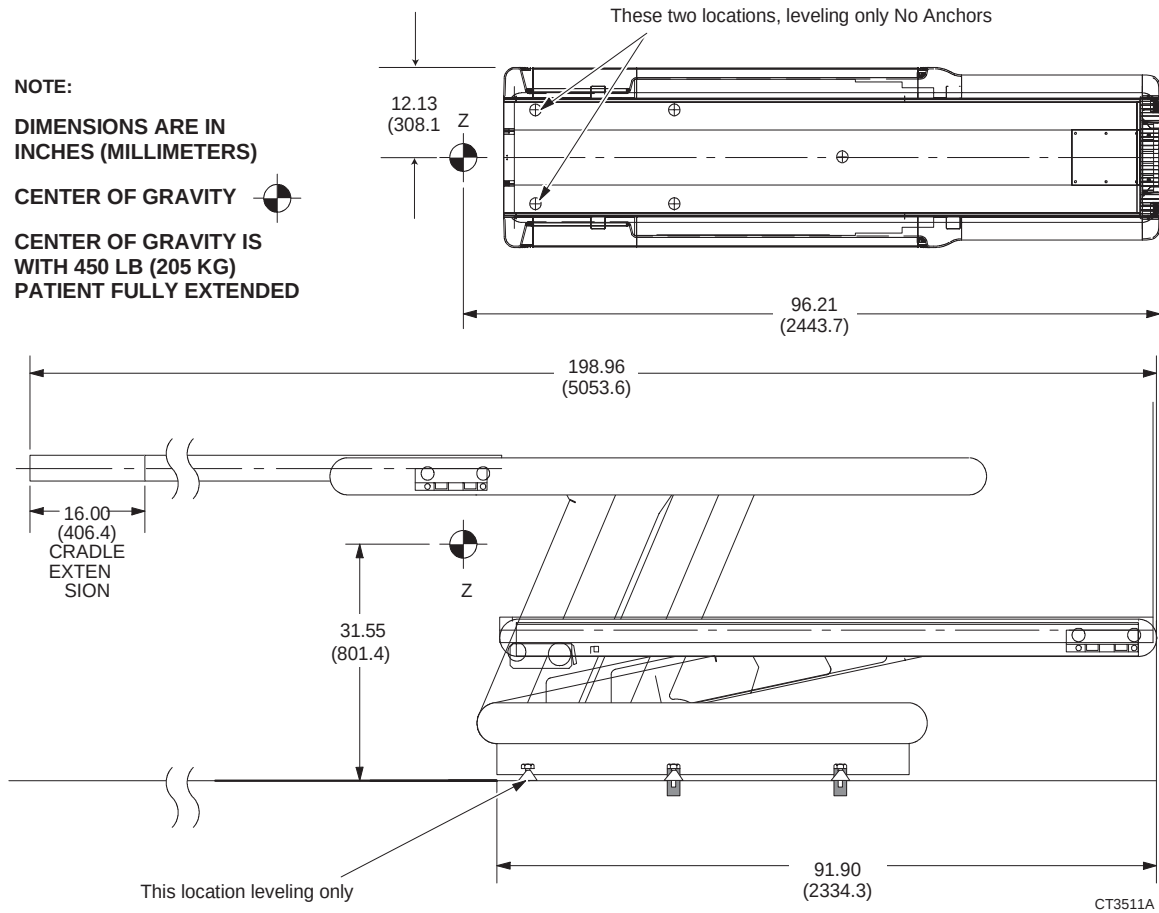


Figure 6-4 Patient Table (CT1)

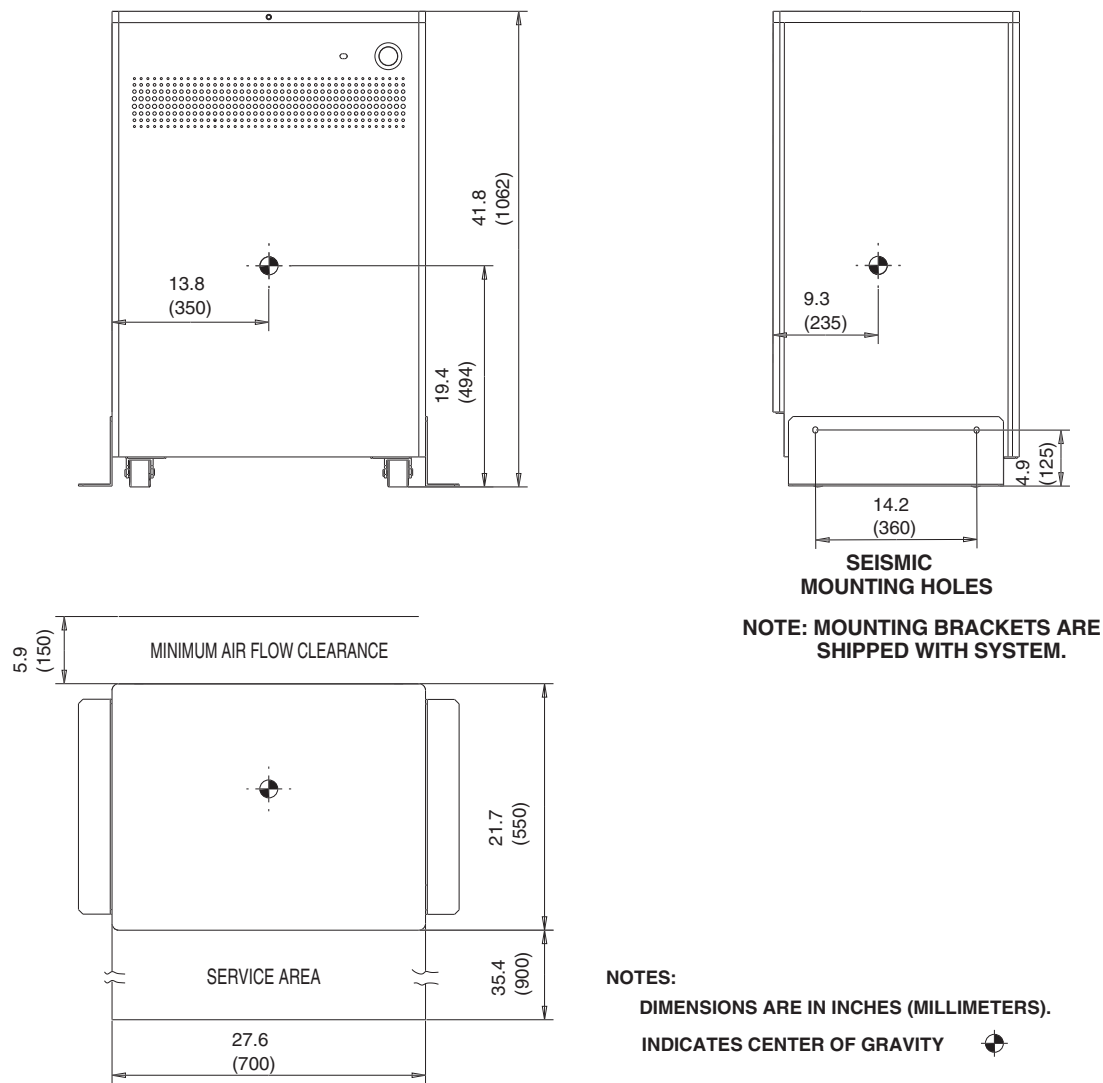


Figure 6-5 Power Distribution Unit (NGPDU)



Chapter 7

Delivery Data



WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0 Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers, which include dollies, skids and boxes without skids.

Section 2.0 Delivery/Shipping Requirements & Considerations

The LightSpeed 5.X System is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping or hoisting.

The Gantry, Console, Table and PDU must NEVER be dropped. A drop from a height greater than one half inch ($\frac{1}{2}$ ") may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the Gantry, Dock to Dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving Gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the Gantry at the slowest reasonable rate. See [Figure 7-1](#).

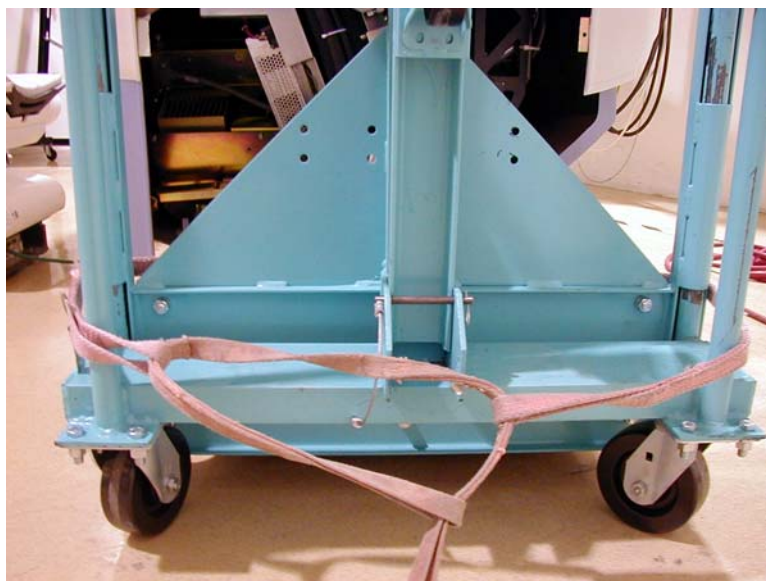


Figure 7-1 Gantry Strap Location

The LightSpeed 5.X System—including Gantry, Console, Table and PDU—is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the Console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive or octane failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. Nor should the Gantry be hoisted. The Gantry should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the Gantry.

Protection for flooring along the move path (from dock to scan room) is advised.

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

LightSpeed 5.X systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- Increased installation time
- Decreased installation quality
- Increased scanner downtime, due to increased service calls

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 5.X CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0 Storage Requirements

SHORT TERM STORAGE (LESS THAN 6 MONTHS)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 40° to +80° F (4° to +27° C).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum relative humidity rate of change is 5%/hr.
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.)
- Air pressure should be between 700hPa and 1060hPa.



NOTICE Between delivery is considered short-term storage.
Van storage must meet the same specifications as above.

LONG TERM STORAGE (6 MONTHS OR MORE)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 50° to +90° F (10° to +32° C).
- Maintain relative humidity (non-condensing) between 20% and 70%.



Figure 7-2 Package Symbols (Storage)

Section 5.0 Extreme Temperature Transportation and Deliveries

Extreme temperatures should be avoided during system transportation and delivery.

Extreme temperatures are defined as below Zero degrees Fahrenheit (-18° C) or above 120 degrees Fahrenheit (49° C), without humidity control.

Section 6.0

System Transportation

When transporting the CT system, ensure that the system is not exposed, for an extended period of time, to temperatures or humidity outside of the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 0% to 80%



NOTICE

Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

Section 7.0

Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies. See [Figure 7-3](#). Two side rails are bolted to the dollies to stabilize dollies and protect gantry. Use dolly elevating casters to lift gantry off its base and roll it into position.



Figure 7-3 Gantry with Shipping Dollies and Side Rails

Door Openings. Clear door openings for moving equipment into building must be 42 X 82 in. (1067 X 2083 mm) minimum. 8 ft. (2439 mm) corridor width is helpful.

Elevator requirements. To move gantry from receiving location to scanning room, consider elevator capacity and size. By removing side rails and one dolly after gantry is placed in elevator, gantry width/length and elevator depth requirements are reduced. Contact a representative of elevator manufacturer if gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH IN (CM)	WIDTH IN (CM)	HEIGHT IN (CM)
Dollies On, Side Rails On	114 (290)	51 (129)	79 (200)
Dollies On, Side Rails Removed	114 (290)	39.4 (100)	79 (200)
Dollies Off, Covers Off	81 (206)	34 (86)	73 (185)

Table 7-1 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for gantry with covers and dollies attached, but side rails removed, is 40 inches (101.6 cm).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Typically, dollies are used on the gantry, table, and console for domestic shipments. Once the gantry, table and console are installed at the site, return dollies to GE using the shipping document located in Box #1.

Dollies can be purchased for international shipments (B7850LD) to be used at the customer site. After the system has been removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be retained to the local GE office or warehouse.

Zero clearance dollies are available (P/N 2128404).

Section 8.0

Table Considerations

The table is shipped without side covers installed. Covers are shipped in four separate boxes. The table is mounted between two dollies.

Table shipping dimensions are 101" (2565 mm) long, 32" (81 mm) wide, and 47" (119 mm) high.

Section 9.0

Console Considerations

The console is shipped without the keyboard table installed. The keyboard table is shipped with the console.

- The console is shipped on a skid equipped with ramps for unloading.
- Do not remove the console from the shipping skid until it is in the CT equipment room.
- Console shipping dimensions (on the shipping skid) are 40" (1016 mm) deep, 53" (1346 mm) wide, and 41" (1041 mm) high.

Note: The dimensions of the console alone (as shipped) are 34.95" (888mm) deep, 48.74" (1238mm) wide, and 29.5" (750mm) high.



NOTICE Do not lift the console by the monitor table top to remove from the shipping skid.



Figure 7-4 Console ready to be unloaded from shipping skid.

Chapter 8

Power Requirements

Section 1.0

Introduction

Note: Power requirements for LightSpeed 5.X differ from those for LightSpeed 4.X and previous systems.

The power distribution unit (PDU) supplied with the LightSpeed 5.0 system transforms and distributes power to all system components. The PDU is the only power entry point required to operate system.

To minimize voltage regulation effects, power wiring between the facility main distribution panel and the PDU should be kept as short as possible.

When routing the power wiring all three phase wires and ground must be run in the same conduit or raceway duct. Power wires should be routed separately from system control and signal cables, using a separate conduit or trough in raceway duct.

Metallic conduit, floor duct or surface raceway may be used for running cables, depending upon local codes and practices. However, cable passageways should be large enough to install any cable with all other cables already installed. Use of non-metallic conduit is not recommended.

Section 2.0

System Input Power

2.1 Power Source Configuration

The LightSpeed 5.0 CT Scanner is designed to operate on a three-phase, four-wire delta or wye power source. Although a solidly grounded wye source is preferred, an impedance grounded wye source may be used. The neutral wire does not need to be run to the system, i.e., four-wire connection.

If the only power source available is a delta configuration the customer must provide a wye connected grounding transformer or autotransformer at the source. Corner or mid-phase grounded delta sources are NOT recommended.

A dedicated feeder from the nearest Main Distribution Panel (MDP) should be used to supply power to the scanner. In accordance with the National Electric Code (U.S.) and similar applicable national and local codes, a protective disconnect device must be provided in the power line supplying the PDU. It must be located within 32 ft (10 m) of the PDU, visible to PDU service personnel, and must have "lockout / tagout" provisions. This disconnect device is identified as "A1" in the interconnection schematic diagrams.

2.2 Rating

The LightSpeed 5.0 system operates on three-phase power meeting the following specifications.

Voltage	200 to 480 VAC
Capacity	90 kVA
Frequency	50 or 60 Hz +/- 3 Hz

- Maximum power demand = 90 kVA @ 0.85 PF at a selected technique of 140 kV, 380 mA.
- Average (continuous) power demand at maximum duty cycle = 20 kVA.

The "A1" disconnect device referenced above must provide overcurrent protection for the system and facilitate multi-point remote "Emergency Off" control of system power. A disconnect utilizing undervoltage release control is preferred over shunt trip devices. The rating of the "A1" disconnect device depends on the nominal line voltage at the site. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for minimum rating requirements and suggested disconnect devices.



NOTICE The electrical rating is described on the system rating label attached on the gantry; not on the PDU.

2.3 Regulation

Total load regulation as measured at the PDU input terminals must not exceed 6%. The capacity of the facility transformer and size & length of feeder wires directly affect the load regulation presented to the system. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for recommended single-unit installation specifics.

2.4 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.5 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in Table 8-1. Maximum transient voltages should be limited to 1500V peak.

2.6 Grounding

Metal conduit, raceway or the armor of armored cable used to power the system should be bonded to the PDU cabinet. However, in addition to such mechanical grounding, a dedicated 1/0 (55 mm²) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.

Note: The shield or armor of armored cable is not sufficient for this purpose.

The ground wire should be bonded to intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0

Recommended Power Distribution System

A dedicated feeder run from the facility main isolation transformer is recommended to power the LightSpeed 5.0 CT scanner. If the scanner must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment that use rapid film changers should be avoided. These changers use a large number of high powered, closely spaced exposures, which may coincide with the CT scan and produce image artifacts.

If a dedicated distribution transformer is provided for the scanner, the minimum recommended transformer size is 112.5 kVA, rated 2.4% regulation at unity power factor. For this configuration, the minimum recommended feeder size and overcurrent protection device based on line voltage is shown in [Table 8-3](#) Minimum Feeder Wire Size.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to the LightSpeed 5.0 CT scanner meet all the requirements stated in this document.

SYSTEM CHARACTERISTICS:

- Maximum power demand = 90kVA @ 0.85 PF: at a Selected Technique of 140 kV, 380 mA.
- Continuous (average) power demand at maximum duty cycle = 20kVA.
- Maximum allowable total source regulation is 6%.
- Minimum recommended transformer size: 112.5 kVA, with 2.4% rated regulation at unity power factor. Resultant maximum allowable feeder regulation is 3.4%.

The nominal line voltage must fall within one of the ranges listed below

Nominal Line Voltage	200	220	240	380	400	420	440	460	480
Hi-Line Limit, +10%	220	242	264	418	440	462	484	506	528
Lo-Line Limit, -10%	180	198	216	342	360	378	396	414	432
Continuous Line Current	58	52	48	30	29	27	26	25	24
Momentary Line Current	260	236	217	137	130	124	118	113	108
Maximum Line Current	289	262	241	152	144	137	131	126	120
Minimum Recommended Circuit Protection Rating	150	150	150	110	110	100	100	90	90

Table 8-1 Nominal Line Voltage

OPTION	CAT NUM
225 kVA Transformer	E4500AW
Isolation Transformer	E4500BC

Table 8-2 Purchasable Options

FEEDER LENGTH (MDATO A1) FEET (METERS)	MINIMUM FEEDER WIRE SIZE, AWG OR MCM (SQ. MM)/ VAC								
	200 VAC	220 VAC	240 VAC	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
50 (15)	1/0 (55)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
100 (30)	2/0 (70)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
150 (46)	4/0 (100)	3/0 (85)	2/0 (70)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
200 (61)	5/0 (125)	4/0 (100)	4/0 (100)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
250 (76)	6/0 (170)	5/0 (125)	5/0 (125)	1 (45)	1 (45)	2 (35)	2 (35)	2 (35)	3 (30)
300 (91)	7/0 (215)	6/0 (170)	5/0 (125)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	2 (35)	2 (35)
350 (107)	8/0 (275)	7/0 (215)	6/0 (170)	2/0 (70)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
400 (122)	8/0 (275)	7/0 (215)	7/0 (215)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)

Table 8-3 Minimum Feeder Wire Size

SUB-FEEDER LENGTH (A1 TO PM) FEET (METERS)	MINIMUM SUB-FEEDER WIRE, AWG OR MCM (SQ. MM)								
	200 VAC	220 VAC	240 VAC	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
32(9.7536)	1/0 (55)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)

Table 8-4 Minimum Feeder Wire Size

- 1.) [Table 8-1](#), [Table 8-3](#), and [Table 8-4](#) above are based on the use of copper wire, rated 75 C and run in steel conduit. Ampacity is determined in accordance with the National Electrical Code (NFPA 70), Table 310-16 (2002)
- 2.) The minimum feeder size is determined by the ampacity of the circuit protection device listed above, except where a larger size is necessary to meet total source regulation limits.
- 3.) A 1/0 (55 sq. mm) ground wire is recommended in all cases.

Section 4.0

Power Source Monitoring

A power audit should be conducted at the customer site prior to installation of the LightSpeed 5.0 CT Scanner. During the audit, a power line analyzer should be used to check the site power for average line voltage, frequency, sags, surges, and transient impulse activity. A minimum ten-day period that includes two weekends should be monitored to encompass several days of normal use. Verify "brown-out" (low voltage) conditions, which may occur during summer months, will not exceed the selected allowable range shown in [Table 7-1](#).

Some analyzer models, which are suitable for power line monitoring, are:

- Dranetz Model 658
- Dranetz Model 656
- BMI 3630

The data and any known site history of previous power problems with other X-ray systems or computer installations should be analyzed and reviewed with your GE Power and Ground representative.

Section 5.0 Ground System

The LightSpeed 5.0 CT Scanner has been designed to use an equal potential grounding system. The required ground system is shown in Figure 7-1. There are three primary grounding points:

- A system power ground point located in the PDU.
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in the patient vicinity are grounded to the reference ground point. For additional information, refer to Electrical Safety Equipment, Direction 46-014505.

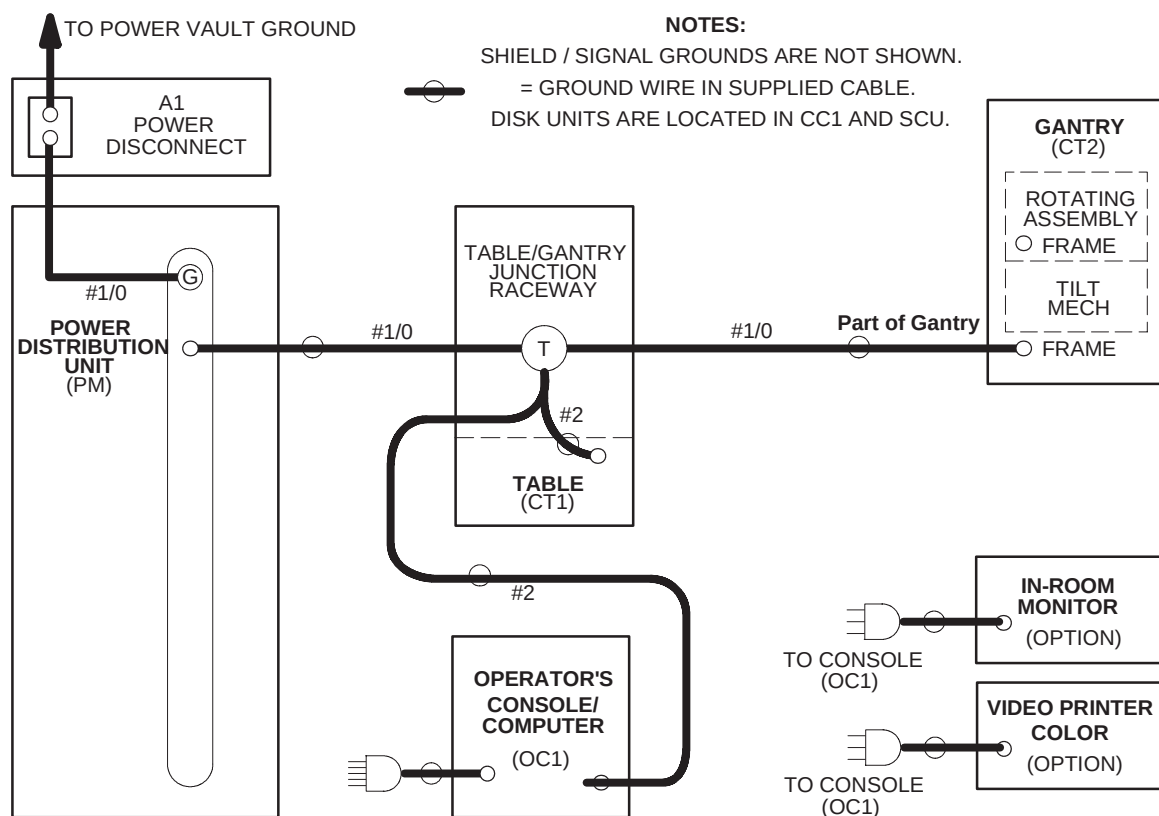


Figure 8-1 System Ground Map

Chapter 9

Interconnection Data

Section 1.0 Introduction

Figure 9-3 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-4 lists customer-installed wiring and supplied cables. Actual length of each run is less than the length of supplied cables to allow for routing inside equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-3 list details for connection to LightSpeed Series equipment, using standard (short) length and non-standard (long) length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

Need for additional junction boxes is minimized by use of either a cable raceway system or a raised computer floor. LightSpeed Series systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

The catalog number of the non-standard (long) length cable set is B7816PH (H-Power Long Cable Set). Order this, if required for your site.

Section 2.0 Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	Contractor supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor supplied
LP	Line printer	Option
OC1	Operator's console/computer	System
PM	Power distribution unit	System
SEO	System emergency off	Contractor supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor supplied
DS	Door Interlock Switch	Contractor supplied
XCVR	Ethernet transceiver	System

Table 9-1 Component Designators

Section 3.0 Interconnect Runs, Wiring and Cables

3.1 GEMS Supplied (Standard Length)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	28 (20)	8.5 (6.1)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	28 (20)	8.5 (6.1)	2343530-2	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (.44) Dia
052	28 (20)	8.5 (6.1)	2343528-2	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	65 (60)	19.8 (18.3)	2343531-2	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
054			n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	43 (35)	13.2 (10.67)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
056	71 (60)	21.7 (18.3)	2371450-4	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	32.5 (20)	9.9 (6.1)	2333152-2	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	71 (60)	21.7 (18.3)	2333150-2	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	68 (60)	20.7 (18.3)	2352714-3	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (.59) Dia
103	68 (60)	20.7 (18.3)	2117848-7	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia
104			n/a	Signal, Gantry to Table		FT-4	300		80		25	22	

Table 9-2 GEMS Supplied Cables (Standard Run) - UL Information

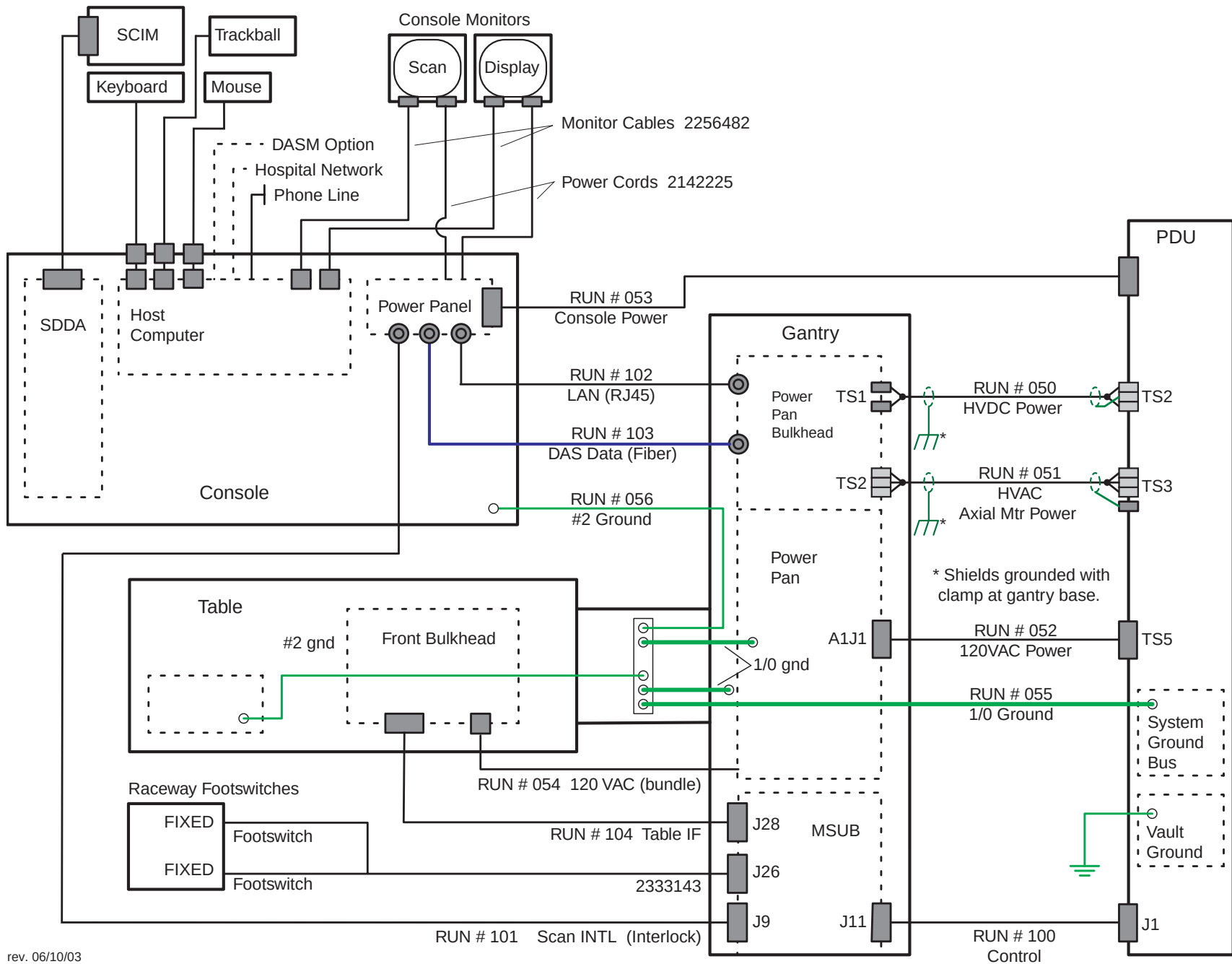


Figure 9-1 System Interconnect Diagram

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3.2 GEMS Supplied (Optional, Long Run)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	63 (55)	19.3 (16.76)	2343529	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	62.5 (55)	19 (16.76)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (.44) Dia
052	60 (55)	18.5 (16.76)	2343528	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	80 (75)	24.5 (22.86)	2343531	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
055	63 (55)	19.3 (16.76)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
056	83 (75)	25.5 (22.86)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2333152	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	83 (75)	25.5 (22.86)	2333150	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2373436-1	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia

Table 9-3 GEMS Supplied Cables (Optional, Long Run) - UL Information

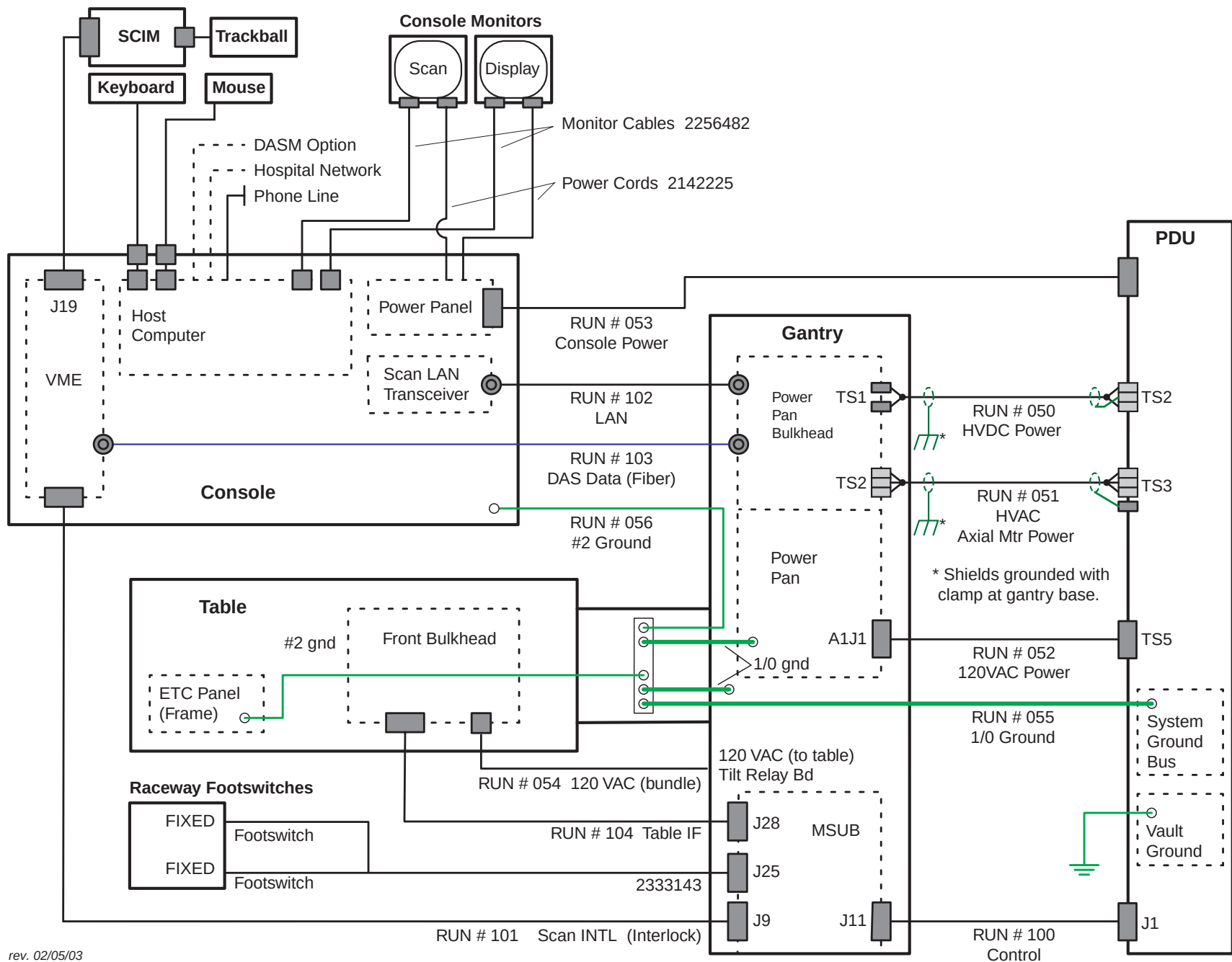


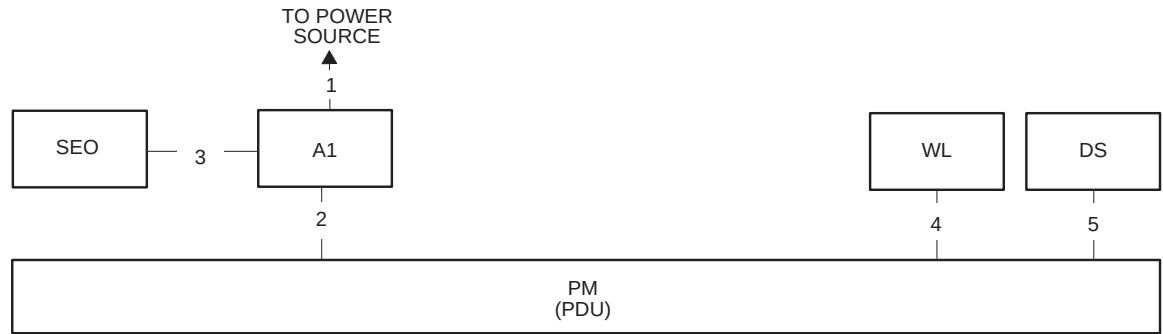
Figure 9-2 System Interconnect Diagram

rev. 02/05/03

3.3 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M.)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*	REFER TO Table 8-4 on page 69 FOR AWG (MM2) WIRE SIZES.								

Table 9-4 Runs 1, 2, 3, 4 and 5 Connections



NOTES:

- 1) Used for remote diagnostics - Option
- 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
- 3) Category 5 cable. Use one of the following patch cords:

CA T Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

- 4) In order to avoid any violation of each National Regulation (NEC in USA, CCC in China, etc.), use of the complied cable/wire is recommended. For China market, China end-user shall purchase the power supply cable that has the CCC mark.

Only one phone connection is required for the system.

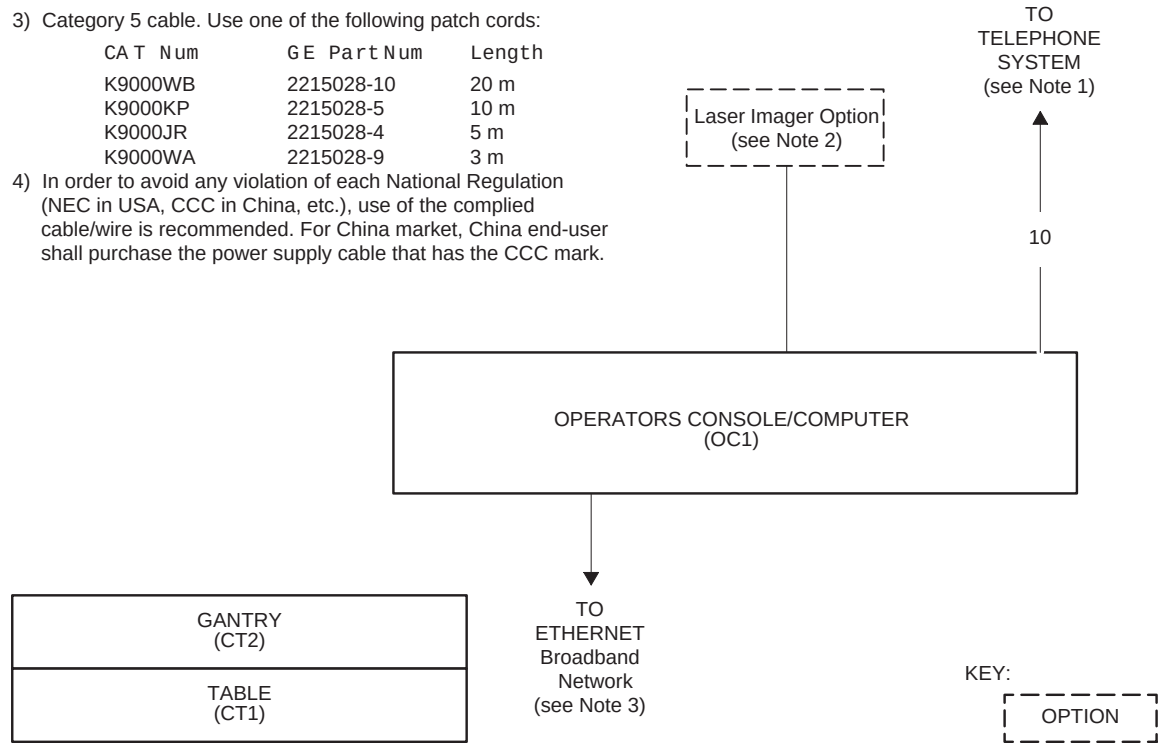


Figure 9-3 Interconnection Runs

Section 4.0

Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1	Fusible Disconnect and Magnetic Contactor	3 Pole, Magnetic Combination Contactor. With control transformer for 220V 50 Hz contactor coil, momentary contact start-stop push-button station, indicator light, multiple neutral block, multiple ground block and dual element fuses.	Source Locally or E4502JP.
ITL	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
MFC	GE LaserCam HQ	See Chapter 10, Section 2.0 , for listings of Camera part numbers and descriptions.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	

Table 9-5 Contractor-Supplied Components

Section 5.0

UPS Interconnect



Figure 9-4 Typical UPS

Section 6.0

Typical Customer Supplied Wiring

6.1 Primary Power Disconnect

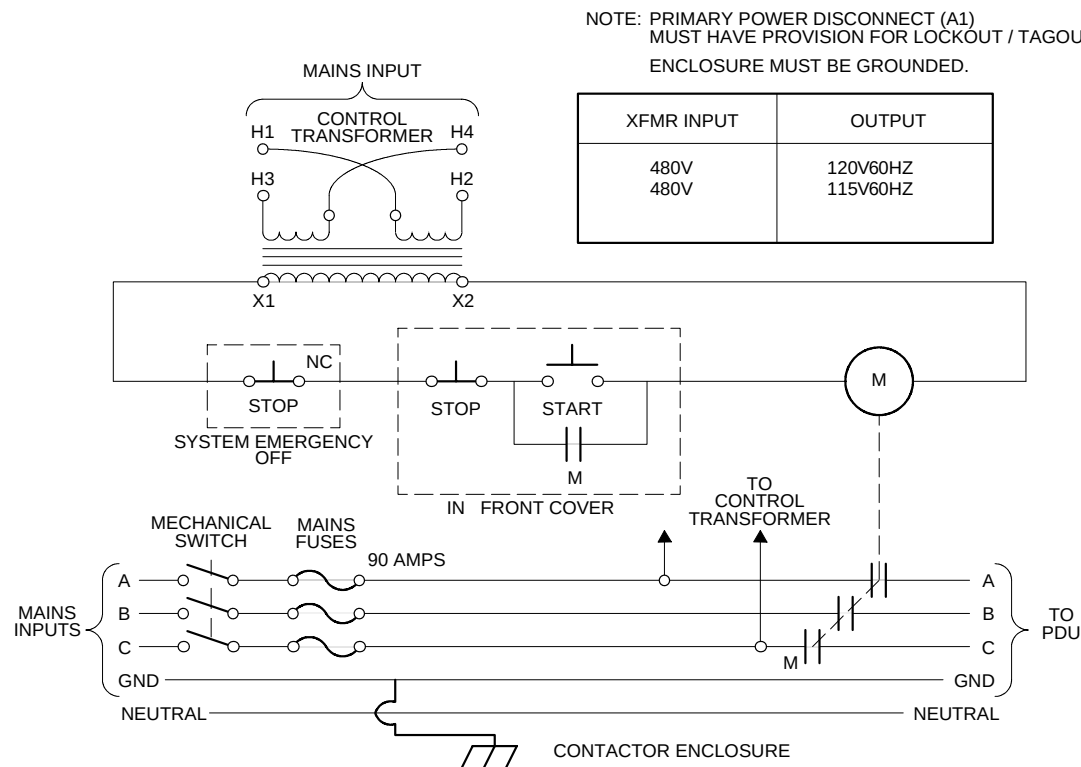


Figure 9-5 Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor



Figure 9-6 Primary Power Disconnect (A1)

6.2 Scan Room Warning Light & Door Interlock

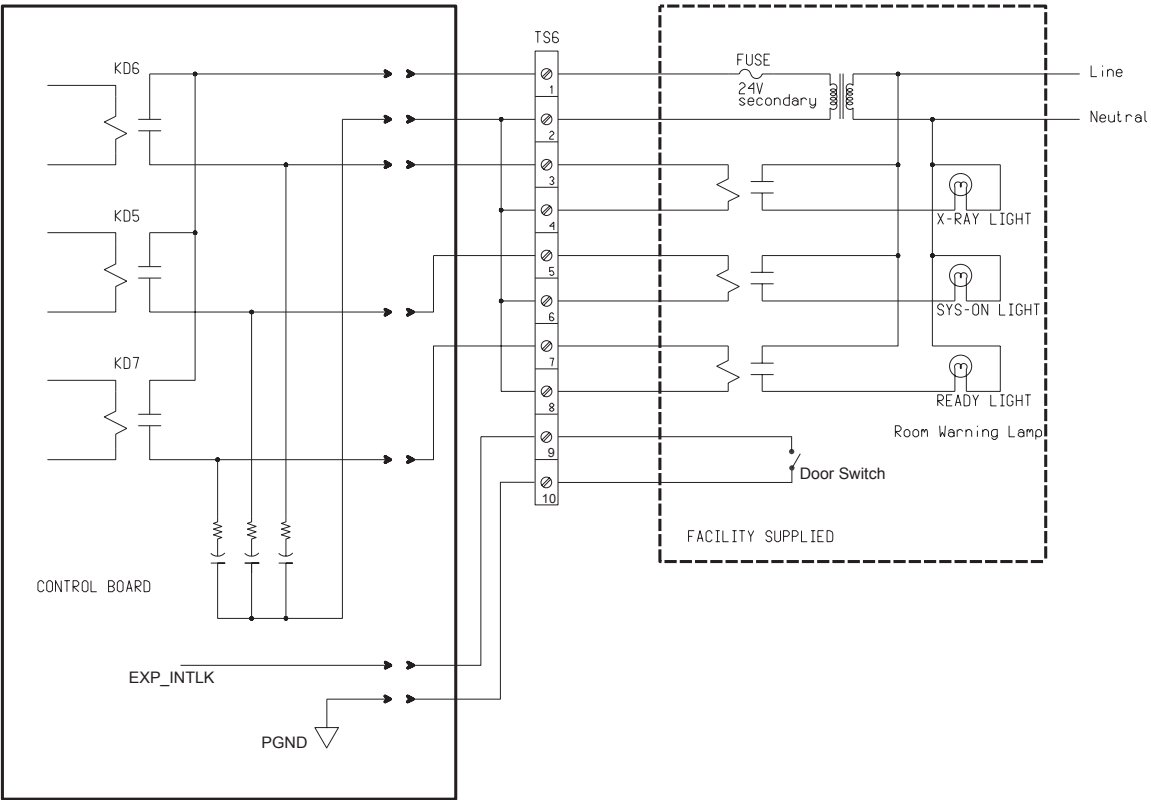


Figure 9-7 Typical TS6 Warning Light & Door Interlock Connections

Appendix A

Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5032	Alternating Current
	335-1	Three-phase Alternating Current
	335-1	Three-phase Alternating Current with neutral conductor
		Direct Current
	417-5019	Protective Earth (Ground)
	348	Attention, consult ACCOMPANYING DOCUMENTS
	417-5008	OFF (Power: disconnection from the mains)
	417-5007	ON (Power: connection to the mains)
		Warning, HIGH VOLTAGE
		Emergency Stop
		Type B

Table 9-6 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5339	X-ray Source Assembly Emitting
	417-5009	Standby
		Start
		Table Set
		Abort
		Intercom
		(on Operator Console) Power On: light on Standby: light off

Table 9-6 Symbols



GE MEDICAL SYSTEMS

**GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.**

**GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE**

**GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPORE**

<http://www.gemedicalsystems.com>